Lecture 2 - Orientations and Symmetries

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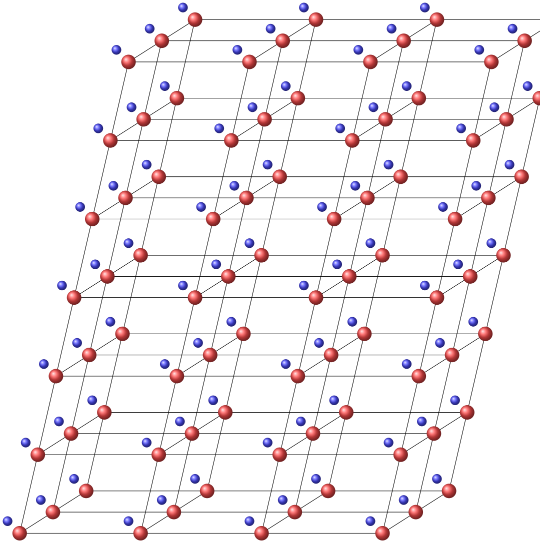
Crystals

Definition

A crystal is an anisotropic, homogenous body consisting of a three-dimensional periodic ordering of atoms, ions or molecules.



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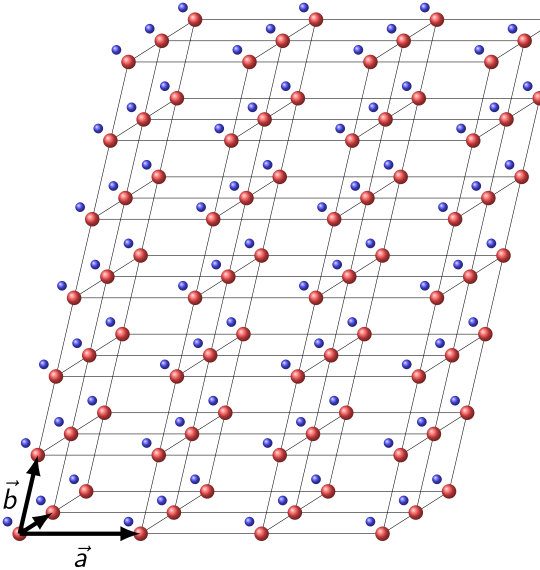
Definition

A crystal is an anisotropic, homogenous body consisting of a three-dimensional periodic ordering of atoms, ions or molecules.

Consequence: There are linearly independent vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ such that the atomic structure is invariant with respect to translations about these vectors.

- The definition of periodicity assumes infinity of the atomic structure.
- The choice of the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ is not unique



The Unit Cell

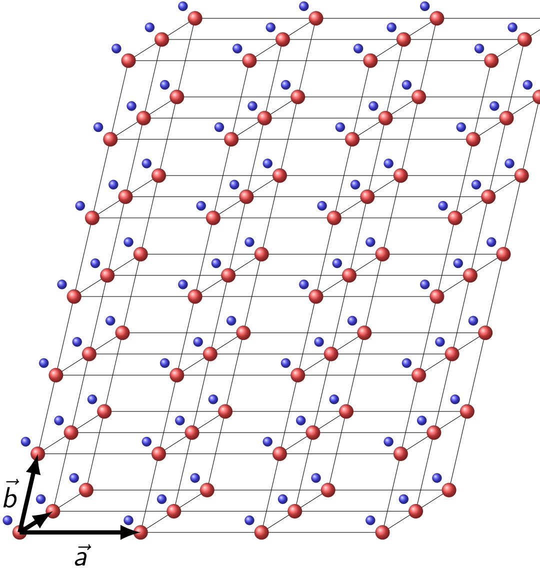
- is the parallelepiped spanned by $\vec{a}$, $\vec{b}$, $\vec{c}$.
- is the smallest region that constitutes a repeating pattern.
- is characterized by the length and angles

$$a = |\vec{a}|, \; b = |\vec{b}|, \; c = |\vec{c}|$$

$$\alpha = \measuredangle(\vec{b}, \vec{c}), \; \beta = \measuredangle(\vec{a}, \vec{c}), \; \gamma = \measuredangle(\vec{b}, \vec{a}).$$

```
abc = [5.2, 5.2, 5.3]
abg = [90 99.25 90] * degree;
cs = crystalSymmetry('1',abc,abg)
```

```
ans = crystalSymmetry
  symmetry           : 1
  a, b, c            : 5.2, 5.2, 5.3
  alpha, beta, gamma: 90, 99.25, 90
```



The Crystal Coordinate Systems

The vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ constitute a right handed but non orthogonal coordinate system.

Lattice points are given by coordinates u, v, w:

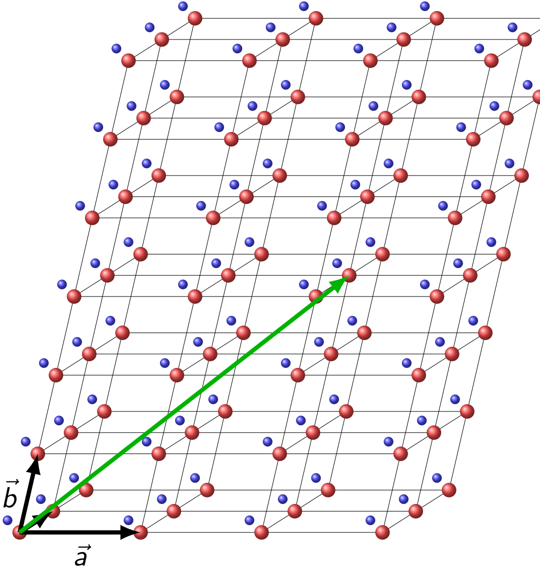
$$\cdot uvw\cdot = \vec{d} = u\vec{a} + v\vec{b} + w\vec{c}.$$

Usage: slip systems, dislocation systems

```
d = Miller(1,1,0,cs,'uvw')
```

```
d = Miller (1)
  u   v   w
  1   1   0
```

Smaller integers u, v, w indicate higher density of lattice points on the line.



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d = Miller(1,1,0,cs,'uvw')
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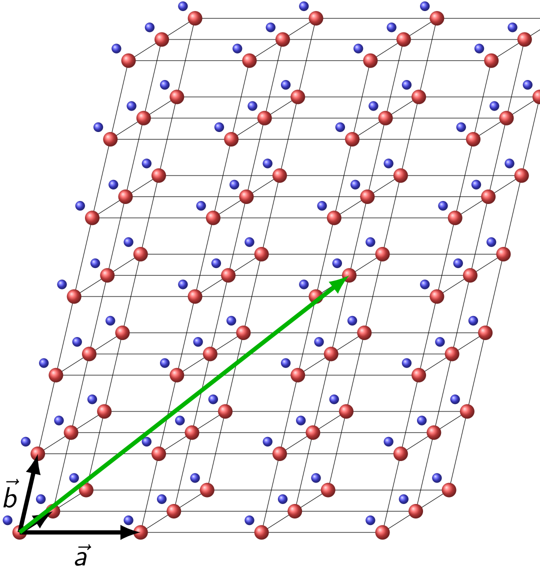
Lattice directions $[uvw]$: all vectors parallel to $\vec{d}$

```
d == [2*d, -d]
```

```
ans =

   1    0
```



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```

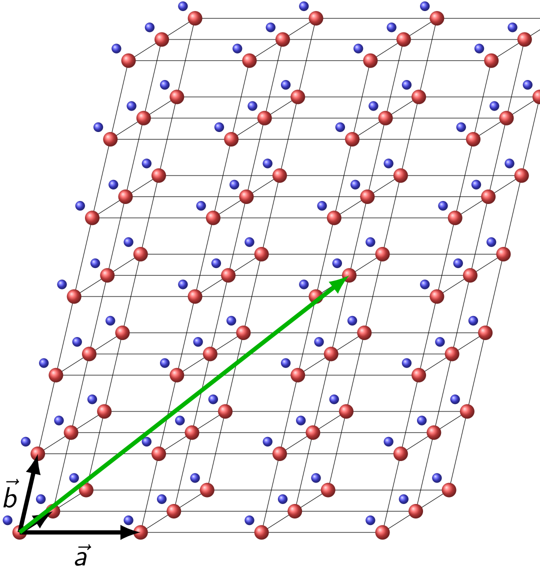
Lattice directions $[uvw]$: all vectors parallel to $\vec{d}$

```
d == [2*d, -d]
```

```
[abs(d), norm(2*d)]
```

```
ans =

    1.4142    2.8284
```



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```
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```

```
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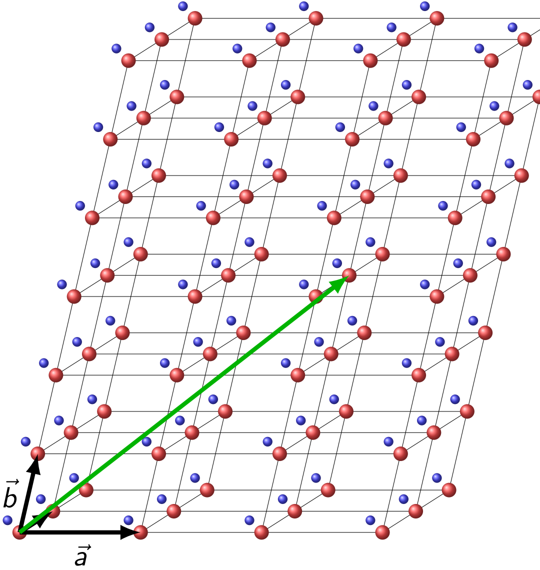
```

Examples: $[100]$, $[1\bar{1}0] = [2\bar{2}0] \neq [\bar{1}10]$



The Reciprocal Coordinate System

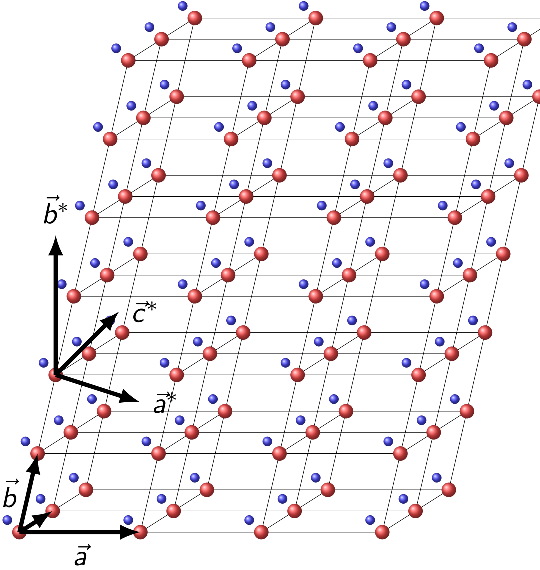
The reciprocal axes are orthogonal to $\vec{a}, \vec{b}, \vec{c}$,

$$\vec{a}^* = \frac{\vec{b} \times \vec{c}}{V}, \; \vec{b}^* = \frac{\vec{c} \times \vec{a}}{V}, \; \vec{c}^* = \frac{\vec{a} \times \vec{b}}{V}$$

- $V = \vec{a} \cdot (\vec{b} \times \vec{c})$ is the volume of the unit cell
- $\vec{a} \cdot \vec{a}^* = 1, \; \vec{b} \cdot \vec{b}^* = 1, \; \vec{c} \cdot \vec{c}^* = 1$
- units of $\vec{a}^*, \vec{b}^*, \vec{c}^*$ are reciprocal to $\vec{a}, \vec{b}, \vec{c}$
- for cubic lattices $\vec{a} || \vec{a}^*, \; \vec{b} || \vec{b}^*, \; \vec{c} || \vec{c}^*$,

Directions with respect to the $\vec{a}^*, \vec{b}^*, \vec{c}^*$ are given by integers coordinates h, k, ℓ

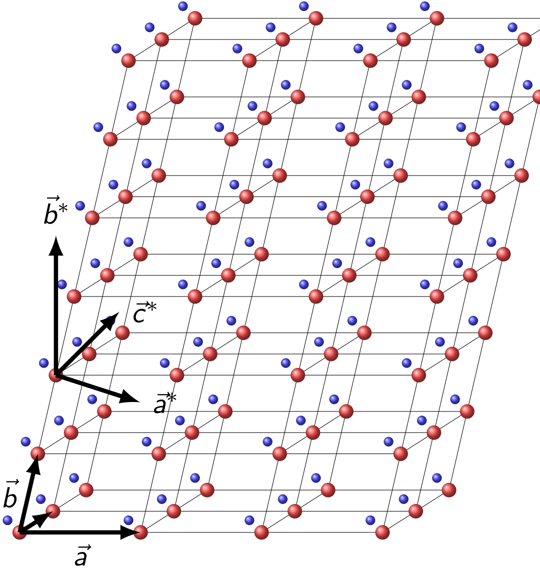
$$\vec{n} = h\vec{a}^* + k\vec{b}^* + \ell c^*$$



The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs,'hkl')
```

```
d = Miller (1)
  h  k  l
  1  1  0
```

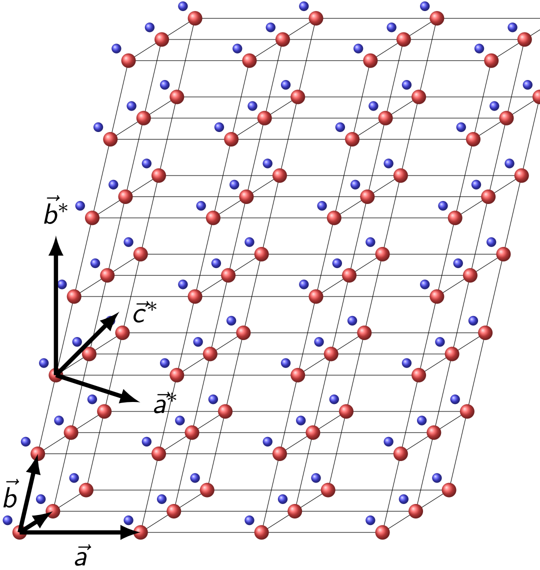


The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs,'hkl')
```

```
Miller({1,1,0},{0,1,1},cs,'hkl')
```

```
ans = Miller (1)
  h   k   l
  1   1   0
```

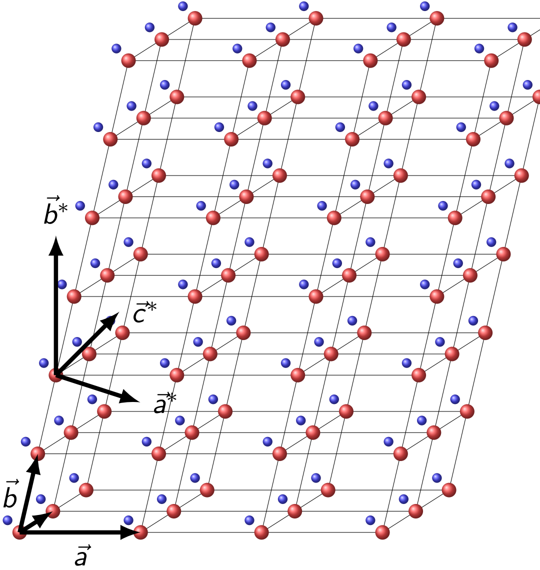


The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs,'hkl')
```

```
Miller({1,1,0},{0,1,1},cs,'hkl')
```

```
rec = [cs.aAxisRec,cs.bAxisRec,...
       cs.cAxisRec]
```



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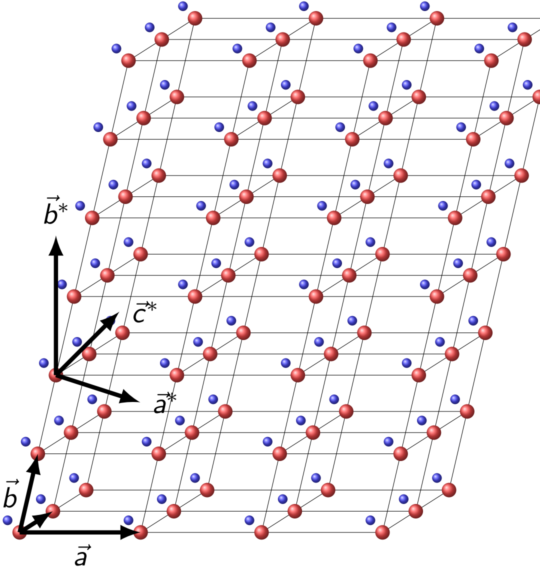
```
rec = [cs.aAxisRec,cs.bAxisRec,...
       cs.cAxisRec]
```

```
dot(rec,cs.aAxis)
```

```
ans =

     1     0     0
```



The Reciprocal Coordinate System

```
n = Miller(1,1,0,cs, 'hkl')
```

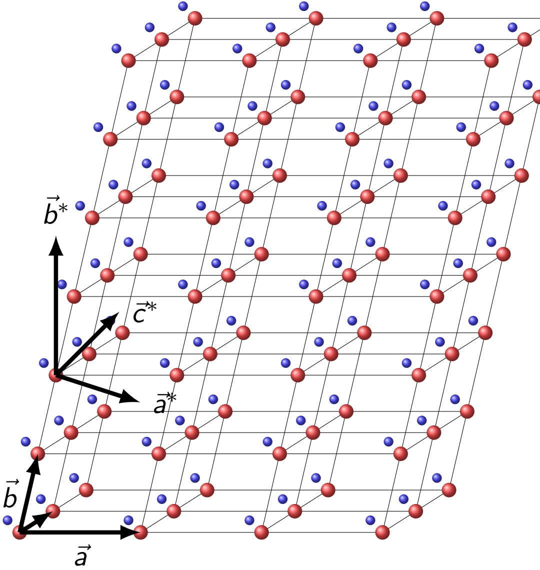
```
Miller({1,1,0},{0,1,1},cs, 'hkl')
```

```
rec = [cs.aAxisRec, cs.bAxisRec, ...
       cs.cAxisRec]
```

```
dot(rec, cs.aAxis)
```

```
cross(cs.aAxisRec, cs.bAxisRec)
```

```
ans = Miller (1)
  u   v   w
  0   0   1
```



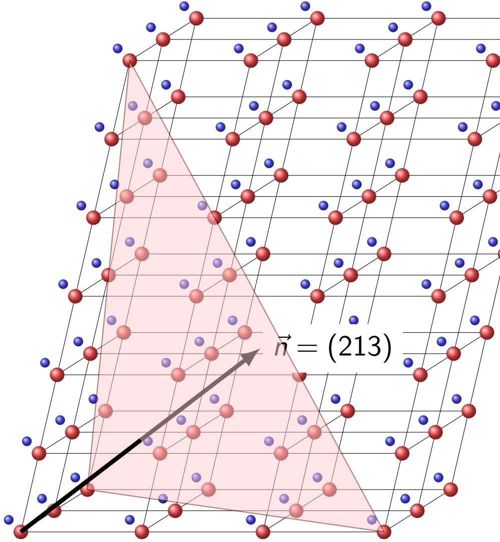
Crystal Lattice Planes

The plane P normal to $\vec{n} = h\vec{a}^* + k\vec{b}^* + \ell c^*$ is

$$P = \{\vec{x} \mid \vec{x} \cdot \vec{n} = d\} = (hk\ell).$$

It intersects the $\vec{a}$-axis in $\cdot u00 \cdot$ if

$$d = u\vec{a} \cdot n = u\vec{a} \cdot (h\vec{a}^* + k\vec{b}^* + \ell c^*) = uh.$$



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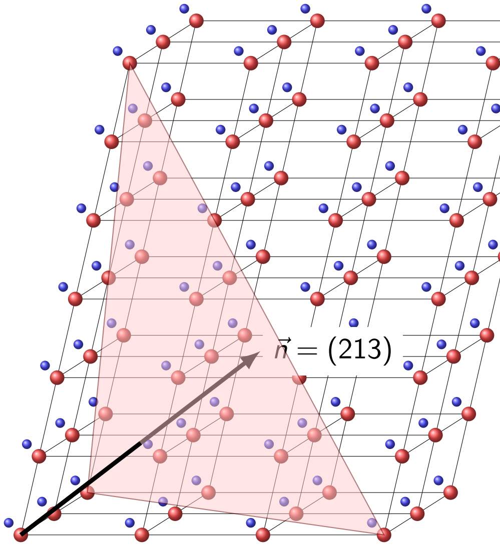
Hence, $(hk\ell)$ intersects $\vec{a}$, $\vec{b}$ and $\vec{c}$ in $\frac{d}{h}$, $\frac{d}{k}$, $\frac{d}{\ell}$.



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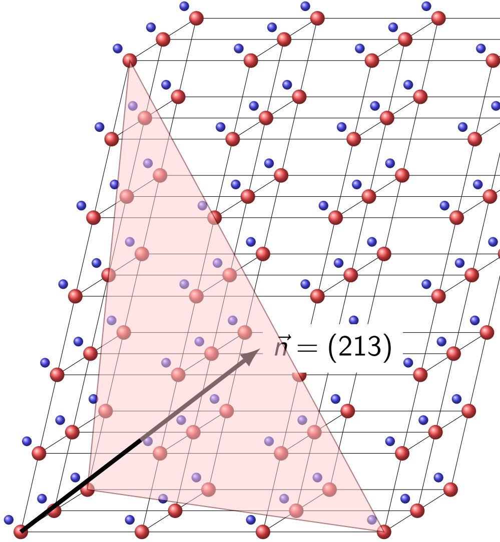
Hence, $(hk\ell)$ intersects $\vec{a}$, $\vec{b}$ and $\vec{c}$ in $\frac{d}{h}$, $\frac{d}{k}$, $\frac{d}{\ell}$.

Given axes intersections $m, n, p \in \mathbb{N}$. The corresponding normal vector is

$$\vec{n} = \frac{d}{m}\vec{a}^* + \frac{d}{n}\vec{b}^* + \frac{d}{p}\vec{c}^*$$

with $d = \mathrm{LCM}(m, n, p)$.



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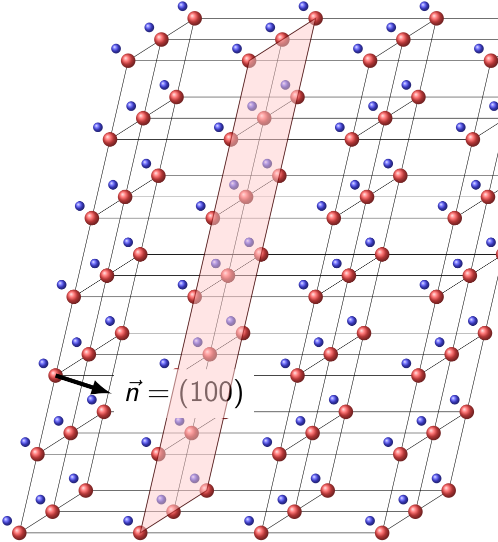
$k = 0 \iff \frac{1}{0} = \infty \iff (hkl)$ is parallel to $\vec{b}$



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It intersects the $\vec{a}$-axis in $\cdot u00 \cdot$ if

$$d = u\vec{a} \cdot n = u\vec{a} \cdot (h\vec{a}^* + k\vec{b}^* + \ell c^*) = uh.$$

Hence, $(hk\ell)$ intersects $\vec{a}$, $\vec{b}$ and $\vec{c}$ in $\frac{d}{h}, \frac{d}{k}, \frac{d}{\ell}$.

Given axes intersections $m, n, p \in \mathbb{N}$. The corresponding normal vector is

$$\vec{n} = \frac{d}{m}\vec{a}^* + \frac{d}{n}\vec{b}^* + \frac{d}{p}\vec{c}^*$$

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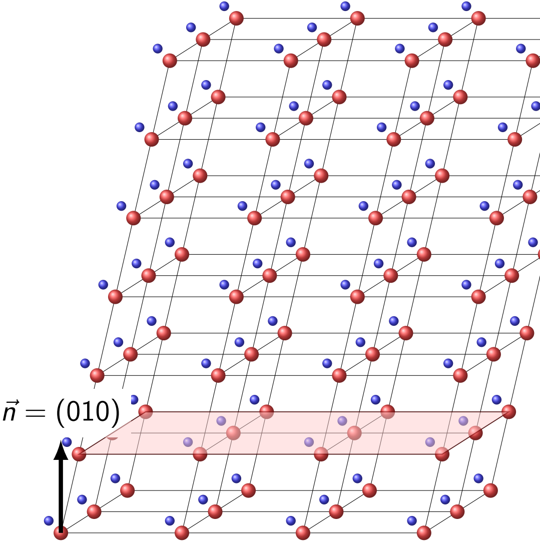
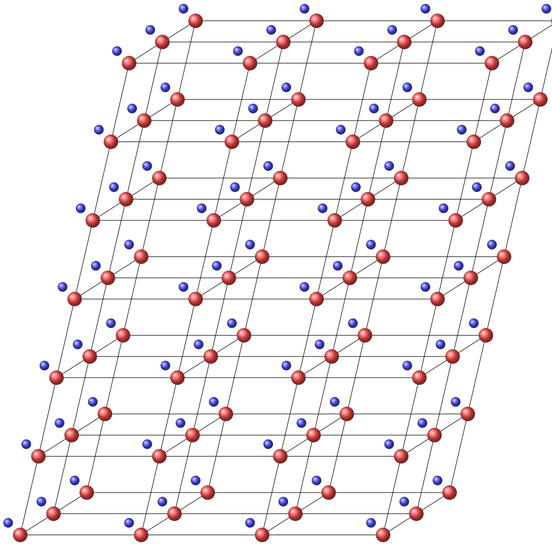


Miller Indices

The lattice plane (123) perpendicular to $\vec{n} = \vec{a}^* + 2\vec{b}^* + 3\vec{c}^*$ which intersects the direct lattice axes $\vec{a}$, $\vec{b}$ and $\vec{c}$ at 1, $\frac{1}{2}$ and $\frac{1}{3}$.

- The coordinates h, k, ℓ are the Miller indices of the plane P.
- planes $(hk\ell)$ parallel to $\vec{a}$ have interception point $\frac{1}{\infty}$ and hence $h = 0$
- planes $(h00)$ perpendicular to $\vec{a}^*$ have interception points $(\frac{d}{h}, \frac{1}{\infty}, \frac{1}{\infty})$



The Zone Equation

A direction $[uvw]$ is parallel to a plane $(hk\ell)$ if

$$hu + kv + \ell w = 0.$$

```
d1 = Miller(1,-1,2,cs,'uvw')
n1 = Miller(1,1,0,cs,'hkl')
dot(d1,n1)
```

```
ans =

   7.7716e-16
```

The Zone Equation

A direction $[uvw]$ is parallel to a plane $(hk\ell)$ if

$$hu + kv + \ell w = 0.$$

```
d1 = Miller(1,-1,2,cs, 'uvw')
n1 = Miller(1,1,0,cs, 'hkl')
dot(d1,n1)
```

```
angle(n1,d2) ./ degree
```

```
ans =

   90.0000
```

The Zone Equation

A direction $[uvw]$ is parallel to a plane $(hk\ell)$ if

$$hu + kv + \ell w = 0.$$

The plane (hkl) spanned by the directions $[u_1 v_1 w_1]$ and $[u_2 v_2 w_2]$:

$$(hkl) = [u_1 v_1 w_1] \times [u_2 v_2 w_2]$$

```
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n1 = Miller(1,1,0,cs,'hkl')
dot(d1,n1)
```

```
angle(n1,d2) ./ degree
```

```
d2 = Miller(1,0,0,cs,'uvw')
cross(d1,d2)
```

```
ans = Miller (1)
  h   k   l
  0   2   1
```

The Zone Equation

A direction $[uvw]$ is parallel to a plane $(hk\ell)$ if

$$hu + kv + \ell w = 0.$$

```
d1 = Miller(1,-1,2,cs,'uvw')
n1 = Miller(1,1,0,cs,'hkl')
dot(d1,n1)
```

```
angle(n1,d2) ./ degree
```

The plane (hkl) spanned by the directions $[u_1 v_1 w_1]$ and $[u_2 v_2 w_2]$:

$$(hkl) = [u_1 v_1 w_1] \times [u_2 v_2 w_2]$$

```
d2 = Miller(1,0,0,cs,'uvw')
cross(d1,d2)
```

The common direction $[uvw]$ of two planes $(h_1 k_1 \ell_1)$ and $(h_2 k_2 \ell_2)$

$$[uvw] = (h_1 k_1 \ell_1) \times (h_2 k_2 \ell_2)$$

```
n2 = Miller(1,0,0,cs,'hkl')
cross(n1,n2)
```

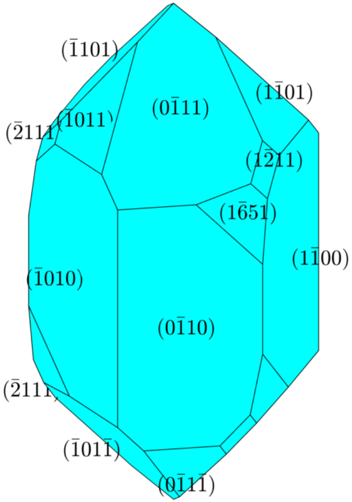
```
ans = Miller (1)
 u   v   w
 0   0  -1
```

Morphology

Crystals are formed by crystallization from a solution or a melt.
the two steps of crystallization: nucleation and growth

- the grow rate is an anisotropic property
- lattice planes with high grow rate get smaller
- lattice planes with low grow rate become larger
- lattice planes $(hk\ell)$ with small Miller indices usually have high density and grow slowly
- the shape of a crystal is formed by its low index lattice planes
- law of constant dihedral angles

```
cs = crystalSymmetry.load('quartz.cif')
cS = crystalShape.quartz;
plot(cS,'colored')
```



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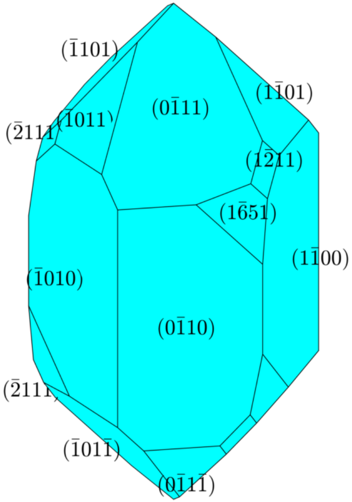

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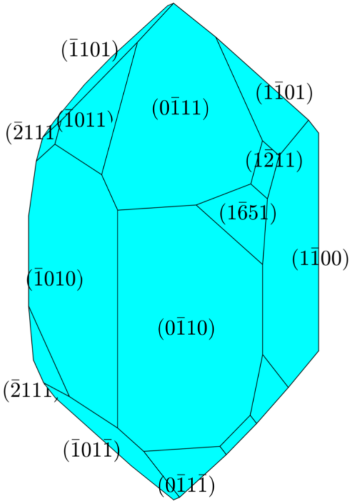

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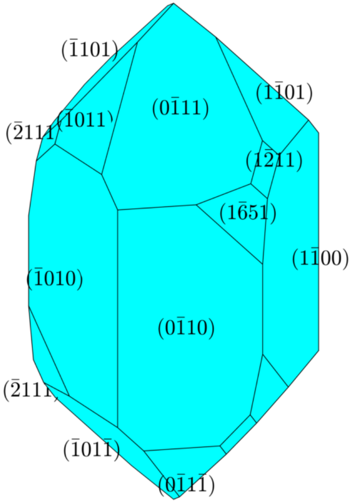

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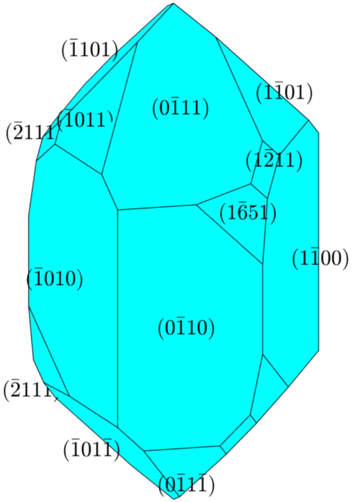

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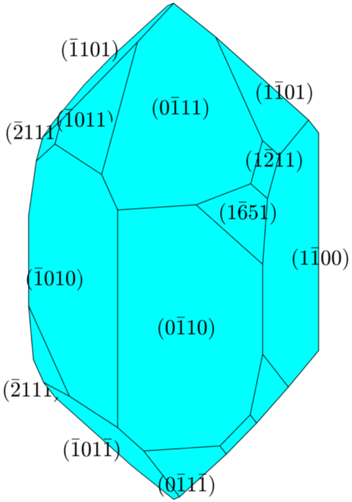

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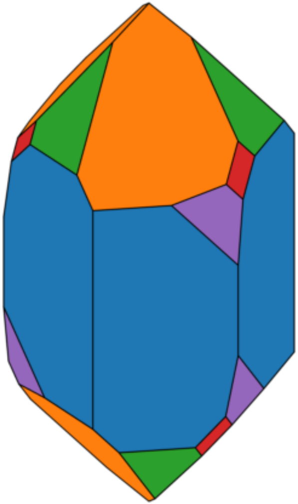
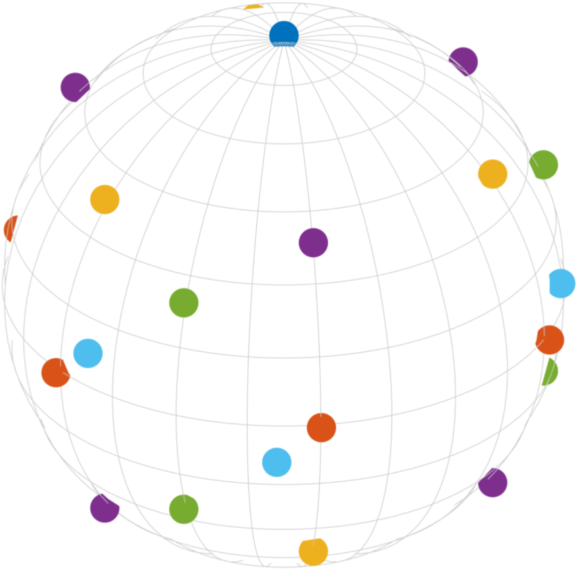


The Crystal Planes of Quartz in a Stereographic Projection



c c-axis:
(0001), $(000\bar{1})$

m hexagonal prism:
$(10\bar{1}0)$, $(01\bar{1}0)$,$(1\bar{1}00)$
$(\bar{1}010)$, $(0\bar{1}10)$,$(\bar{1}100)$

r positive rhomboedron:
$(10\bar{1}1$, $(\bar{1}101)$, $(0\bar{1}11)$

z negative rhombodron:
$(1\bar{1}01)$,$(01\bar{1}1)$,$(\bar{1}011)$

s trigonal bipyramid:
$(11\bar{2}1)$, $(1\bar{2}11)$, $(\bar{2}111)$

x positive trapezohedron:
$(51\bar{6}1)$, $(\bar{6}511)$, $(1\bar{6}51)$

Active vs. Passive Rotations

Active Rotations

An active rotation is a mapping of the Euclidean space onto itself that keeps at least one point and all distances invariant and preserves orientation.

Passive Rotations

A passive rotation is a coordinate transform from one right handed, orthonormal coordinate system into another one.

Improper Rotations

An improper rotation is a rotation that switches between left handed and right handed coordinate systems.

the matrix $\mathbf{M} = \begin{pmatrix} | & | & | \\ \vec{v}_1 & \vec{v}_2 & \vec{v}_3 \\ | & | & | \end{pmatrix}$

rotates

$\vec{e}_1 \mapsto \vec{v}_1,\ \vec{e}_2 \mapsto \vec{v}_2,\ \vec{e}_3 \mapsto \vec{v}_3$

$\mathbf{M}$ transforms coordinates from the reference frame $(\vec{v}_1, \vec{v}_2, \vec{v}_3)$ into the reference frame $(\vec{e}_1, \vec{e}_2, \vec{e}_3)$

$\tilde{\mathbf{M}} = -\mathbf{M}$ additionally mirrors all vectors at the origin

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
```

```
r = rotation

  Bunge Euler angles in degree
  phi1  Phi  phi2  Inv.
     0   90     0     0
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
```

by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
```

```
r = rotation

  Bunge Euler angles in degree
  phi1   Phi   phi2   Inv.
    90     0      0     0
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
```

by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
```

by a rotation matrix

```
M = [1 0 0; 0 0 -1; 0 1 0]
r = rotation.byMatrix(M)
```

```
r = rotation

  Bunge Euler angles in degree
  phi1   Phi   phi2   Inv.
     0    90      0     0
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
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by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
```

by a rotation matrix

```
M = [1 0 0; 0 0 -1; 0 1 0]
r = rotation.byMatrix(M)
```

by pairs of vectors

```
u1 = vector3d.Z; v1 = u1
u2 = vector3d.X; v2 = vector3d.Y
r  = rotation.map(u1,v1,u2,v2)
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
```

by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
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u2 = vector3d.X; v2 = vector3d.Y
r  = rotation.map(u1,v1,u2,v2)
```

```
r  = rotation.fit(u, v)
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
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by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
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u2 = vector3d.X; v2 = vector3d.Y
r  = rotation.map(u1,v1,u2,v2)
```

```
r  = rotation.fit(u, v)
```

import rotations

```
r = rotation.load('file.txt',...
  'ColumnNames',...
  {'phi1','Phi','phi2'})
```

```
r = rotation
  size: 1 x 100
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
```

by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
```

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```

```
r  = rotation.fit(u, v)
```

import rotations

```
r = rotation.load('file.txt',...
  'ColumnNames',...
  {'phi1','Phi','phi2'})
```

random rotations

```
r = rotation.rand(100)
```

```
r = rotation
  size: 1 x 100
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
w = 90*degree
r = rotation.byAxisAngle(v,w)
```

by Euler angles

```
r = rotation.byEuler(0,0,pi/2)
```

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r  = rotation.fit(u, v)
```

import rotations

```
r = rotation.load('file.txt',...
  'ColumnNames',...
  {'phi1','Phi','phi2'})
```

random rotations

```
r = rotation.rand(100)
```

identity and inversion

```
r = rotation.id
r = rotation.inversion
```

```
r = rotation

  Bunge Euler angles in degree
  phi1   Phi  phi2  Inv.
     0     0     0     0
     0     0     0     1
```

Defining Active Rotations

by axis angle

```
v = vector3d(1,0,0)
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```

```
r  = rotation.fit(u, v)
```

import rotations

```
r = rotation.load('file.txt',...
  'ColumnNames',...
  {'phi1','Phi','phi2'})
```

random rotations

```
r = rotation.rand(100)
```

identity and inversion

```
r = rotation.id
r = rotation.inversion
```

a reflextion

```
r = reflection(vector3d.X)
```

```
r = rotation

  Bunge Euler angles in degree
  phi1  Phi  phi2  Inv.
     0  180     0     1
```

Euler Angles

Definition (Euler angles)

Let $\varphi_1, \varphi_2 \in [0, 2\pi]$ and $\Phi \in [0, \pi]$. Then $\varphi_1, \Phi, \varphi_2$ are called the Euler angles of the rotation

$$\mathbf{R}(\varphi_1, \Phi, \varphi_2) = \mathbf{R}_{\vec{Z},\varphi_1}\mathbf{R}_{\vec{X},\Phi}\mathbf{R}_{\vec{Z},\varphi_2}.$$

```
rot = rotation.byEuler(10*degree,20*degree,30*degree)
```

- For every rotation $\mathbf{R}$ there are Euler angles $\varphi_1, \Phi, \varphi_2$ such that $\mathbf{R} = \mathbf{R}(\varphi_1, \Phi, \varphi_2)$.
- For specific rotations **R** the Euler angles are not unique, e.g.

$$\mathbf{R}_{\vec{z},\omega} = \mathbf{R}(\varphi_1, 0, \omega - \varphi_1)$$

- Euler angles are the most common way to specify and visualize rotations in texture analysis.
- The ambiguity makes visualization with respect to Euler angles hard to interpret.

Operations with Rotations

vector rotation

```
v = rot .* u
```

concatenation

```
rot = rot1 .* rot2
v = rot .* u
v = rot1 * (rot2 * u)
```

inverse of a rotation / misrotation

```
inv(rot1)
inv(rot1) .* rot2
```

basic statistics

```
mean(rot)
mean(rot,'weights',w)
std(rot)

unique(rot)
```

convert to Euler angles

```
[phi1,Phi,phi2] = Euler(rot)
```

axis / angle

```
rot.axis, rot.angle
angle(rot1,rot2)
```

convert to matrix

```
rot.matrix
```

convert to quaternion

```
[a,b,c,d] = double(rot)
quaternion(rot)
```

convert to rotation vectors

```
rot.Rodrigues
rot.homochoric
rot.cubochoric
```

Visualizing Rotations

Euler angle space

```
rot = rotation.rand
plot(rot)
```

axis angle space

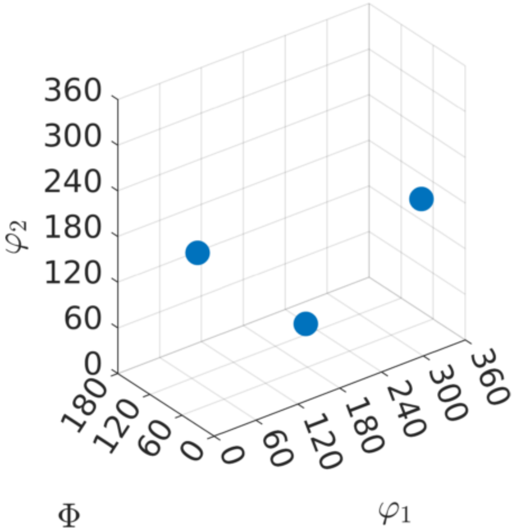
```
plot(rot, 'axisAngle')
```

Euler angle φ_2 sections

```
plotSection(rot, 'phi2')
```

axis angle sections

```
plotSection(rot, 'axisAngle')
```



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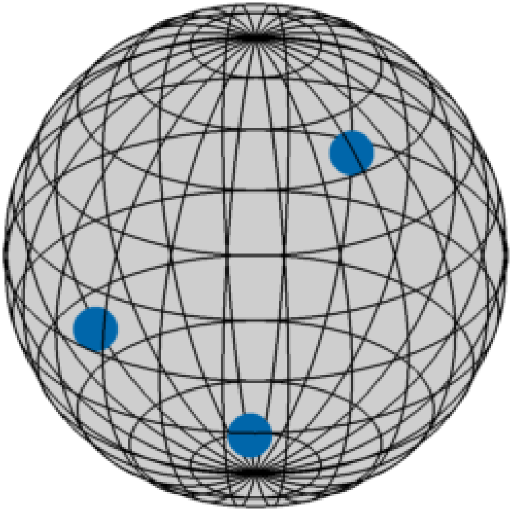

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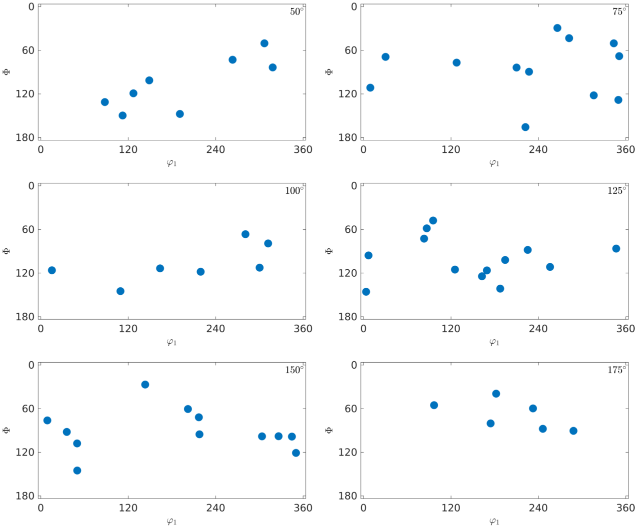

Visualizing Rotations

Euler angle space

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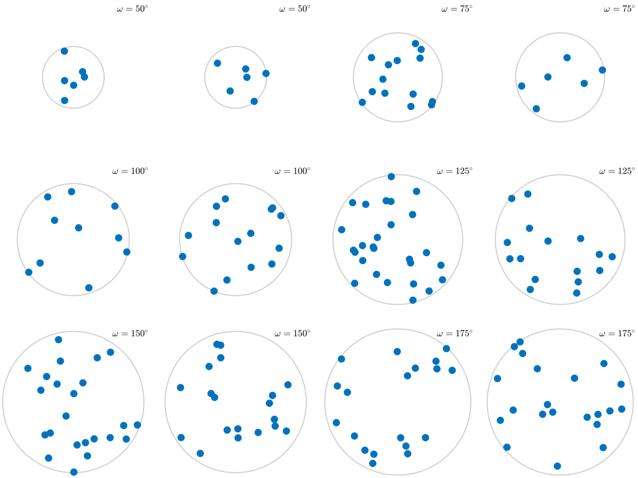


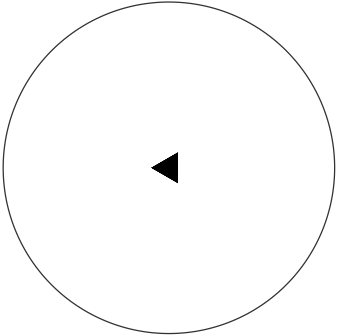
Summary of Rotation Representations

name	notation	space	dimension
matrix	$\mathbf{R}$	$\mathbb{R}^{3\times 3}$	9
Euler angles	$(\varphi_1, \Phi, \varphi_2)$	$[0, 2\pi] \times [0, \pi] \times [0, 2\pi]$	3
quaternion	(q_1, q_2, q_3, q_4)	$\mathbb{S}^4$	4
Rodrigues - Frank	$\tan\frac{\omega}{2}\vec{v}$	$\mathbb{R}^3$	3
axis angle	$\omega\vec{v}$	$\mathbb{B}^3$	3
Miller-Bravais Indices	$(hkl)[uvw]$	$\mathbb{S}^2 \times \mathbb{S}^2$	6

Symmetries

Definition

- A crystal symmetry is a transformation (translation + rotation) that keeps the crystal lattice invariant.
- The group of all crystal symmetries is called space group.
- Their rotational parts form the point group.



The symmetry elements in a point group are generated by

- rotations about 180, 120, 90 or 60 degrees
- reflections
- the inversion

```
Z   = vector3d.Z
rot = rotation.byAxisAngle(Z,120*degree)
cs  = crystalSymmetry.byElements(rot)
```

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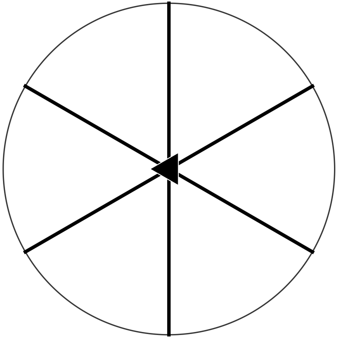

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```
Z   = vector3d.Z
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m   = reflection(vector3d.X)
cs  = crystalSymmetry.byElements([rot, m])
```

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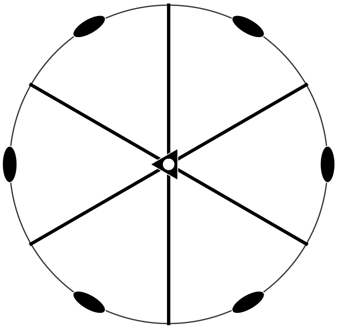

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```

```
cs  = cs.add(rotation.inversion)
```

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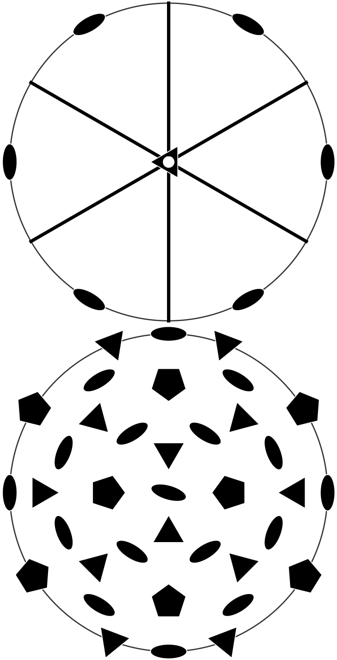
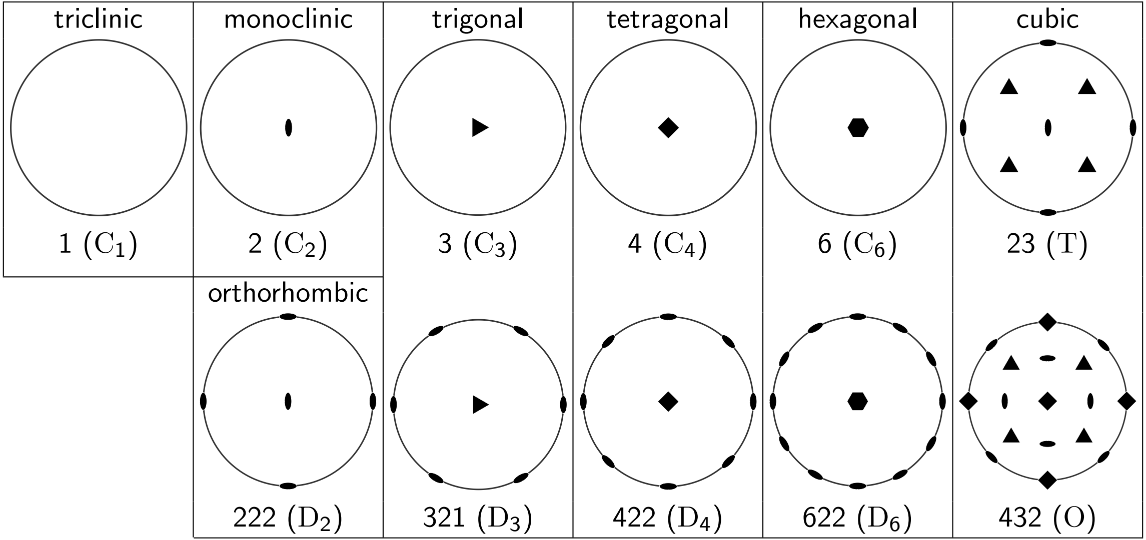
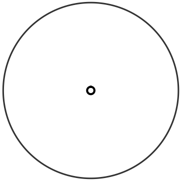
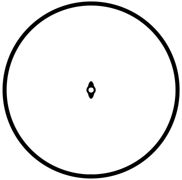
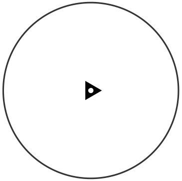
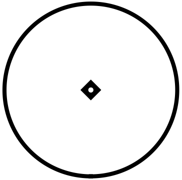
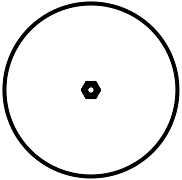
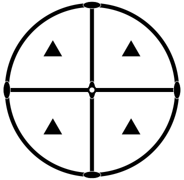
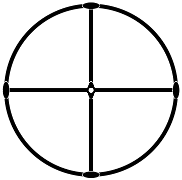
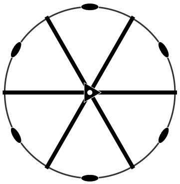
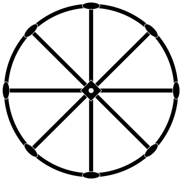
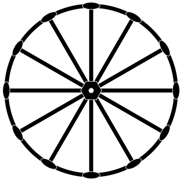
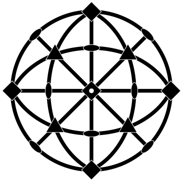


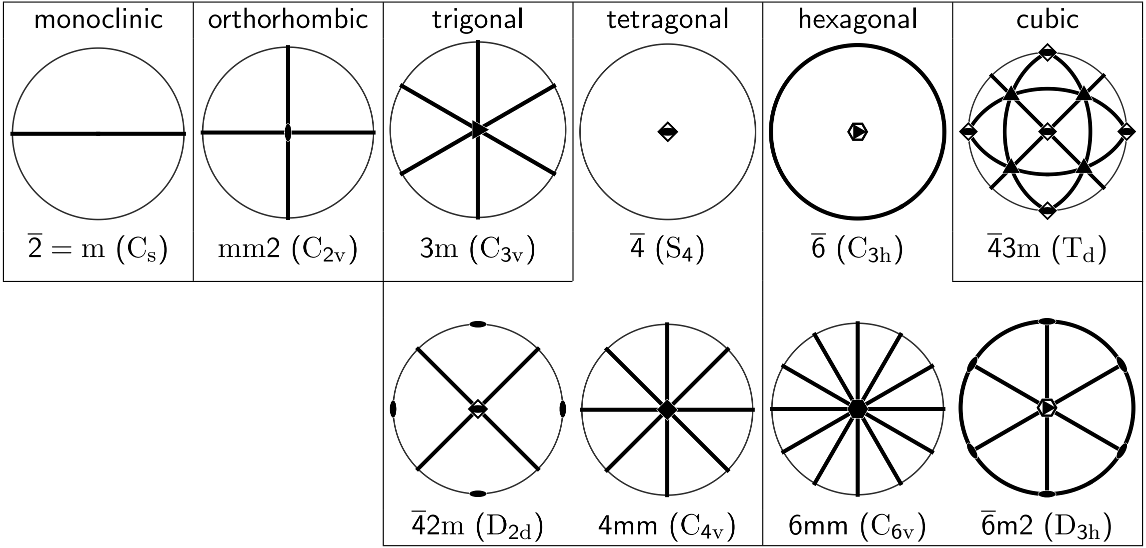
All 11 Enantiomorphic Symmetry Groups



All 11 Laue Symmetry Groups

triclinic	monoclinic	trigonal	tetragonal	hexagonal	cubic
					
$\bar{1}$ (C_i)	$\frac{2}{m}$ (C_{2h})	$\bar{3}$ (C_{3i})	$\frac{4}{m}$ (C_{4h})	$\frac{6}{m}$ (C_{6h})	$m\bar{3} = \frac{2}{m}\bar{3}$ (T_h)
	orthorhombic				
					
	mmm (D_{2h})	$\bar{3}m = \bar{3}\frac{2}{m}$ (D_{3d})	4/mmm (D_{2h})	6/mmm (D_{6h})	$m\bar{3}m$ (O_h)

10 mixed groups



Defining Crystal Symmetries in MTEX

```
cs = crystalSymmetry('m-3m')  % by international symbol
cs = crystalSymmetry('Oh')    % by Schoenflies notation
cs = crystalSymmetry('Fm-3m') % by space group

% import CIF file
cs = crystalSymmetry.load('quartz.cif')

% download from Crystallography Open Database
cs = crystalSymmetry.load('5000036')
```

```
cs.properGroup
cs.Laue
cs.rot
```

```
cs.how2plot.east = cs.bAxis
plot(cs)
```

The Ambiguity of the Crystal Coordinate System

the axes of the crystal coordinate system $\vec{a}$, $\vec{b}$, $\vec{c}$ are always chosen such that

- the translations $\mathbf{T}_{\vec{a}}$, $\mathbf{T}_{\vec{b}}$, $\mathbf{T}_{\vec{c}}$ are symmetry elements of the space group
- $\vec{c}$ is the axis of highest symmetry (except for monoclinic and 23)
- $\vec{a}$ or $\vec{b}$ are aligned with the symmetry axes perpendicular to $\vec{c}$

critical symmetries

monoclinic	alignment of the two fold axis: 211, 121, 112, $m11$, $1m1$, $11m$,
orthorhombic	alignment of the two fold axis: $2mm$, $m2m$, $mm2$
trigonal	alignment of the two fold axis: 321, 312

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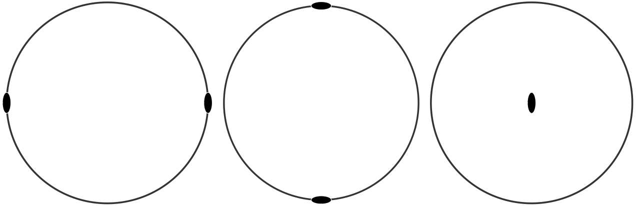

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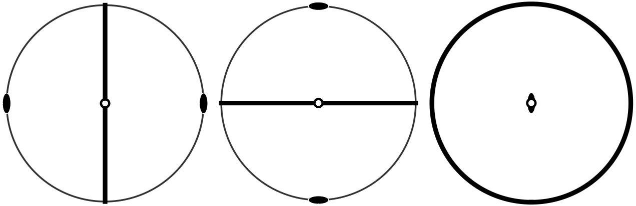

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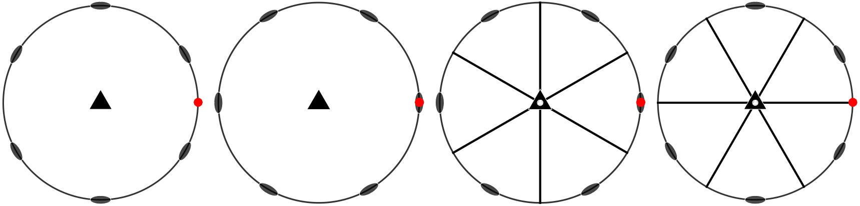

Symmetrically Equivalent Lattice Directions

The crystal axes $\vec{a}$, $\vec{b}$, $\vec{c}$ can be well defined modulo actions of the symmetry group $\mathcal{S}$.

The axes $(\vec{a}, \vec{b}, \vec{c})$ and $(\mathbf{S}\vec{a}, \mathbf{S}\vec{b}, \mathbf{S}\vec{c})$ are physically indistinguishable for all $\mathbf{S} \in \mathcal{S}$

Two lattice directions

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$$\vec{d}_2 = \mathbf{S}\vec{d}_1.$$

- $\langle uvw \rangle$ denotes the set of all lattice directions symmetrically equivalent to $[uvw]$
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For most symmetries the symmetrically equivalent directions comes as permutations of the Miller indices

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Analogously the class of symmetrically equivalent lattice planes $\{hk\ell\}$ is defined as the set of all lattice planes $(h_2 k_2 \ell_2)$ such that there is a symmetry operation $\mathbf{S} \in \mathcal{S}$ with

$$(h_2 k_2 \ell_2) = h_2 \vec{a}^* + k_2 \vec{b}^* + \ell_2 \vec{c}^* = \mathbf{S}(h\vec{a}^* + k\vec{b}^* + \ell\vec{c}^*)$$

reference system	single	symmetrically equivalent
lattice point	$\cdot uvw \cdot$	$: uvw :$
lattice direction	$[uvw]$	$\langle uvw \rangle$
lattice plane	$(hk\ell)$	$\{hkl\}$

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Miller Indices for Trigonal and Hexagonal Symmetries

For trigonal and hexagonal symmetries symmetrically equivalent directions can **not** be determined by permuting the three digit Miller indies.

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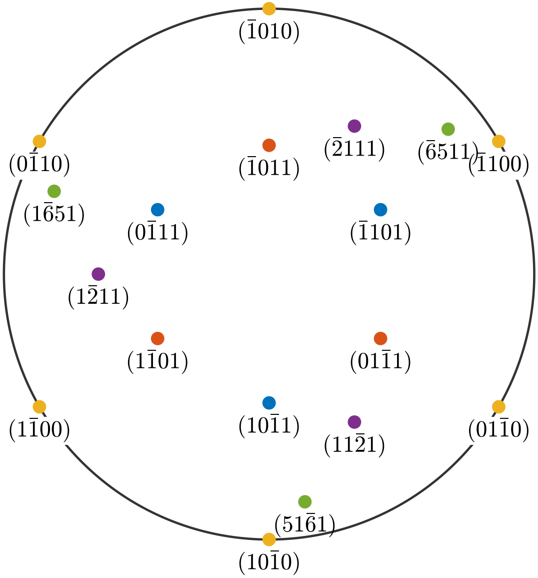
For trigonal and hexagonal symmetries symmetrically equivalent directions can **not** be determined by permuting the three digit Miller indies.

Solution: 4 digit Miller indices

$(HKIL)$ $H = h$, $K = k$, $i = -h - k$, $L = \ell$

$[UVTW]$ $U = 2u - v$, $V = 2v - u$, $T = -u - v$, $W = 3w$



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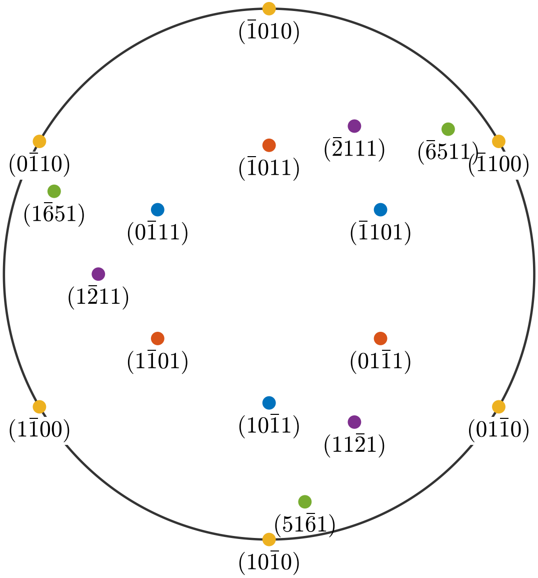

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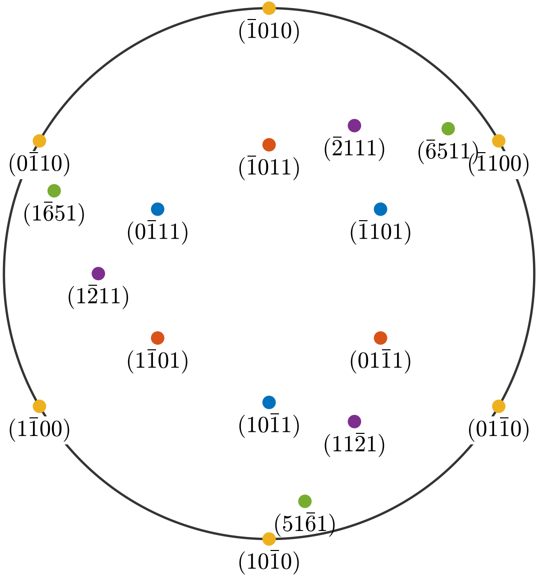


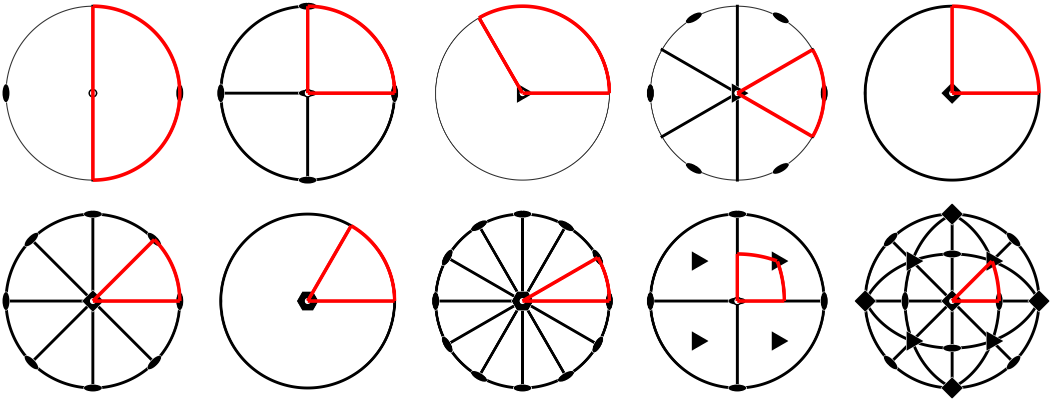
MTEX

```
% define some crystal direction
cs = crystalSymmetry('321',[4.9 4.9 5.4],'mineral','quartz')
h = Miller({1,0,-1,0},{0,0,0,1},cs,'UVTW')

% define some crystal plane
h = Miller({1,0,-1,0},{0,0,0,1},cs,'HKIL')

% find all symmetrically equivalent
hSym = h.symmetrise
unique(h.symmetrise,'noSymmetry')

h.multiplicity

angle(hSym(1),hSym(2))

angle(hSym(1),hSym(2),'noSymmetry')
```

The Fundamental Sector

Definition

The fundamental sector is a spherical region which contains from each class of symmetrically equivalent vectors exactly one.



MTEX

```
% the fundamental sector for a given symmetry
sR = cs.fundamentalSector
plot(sR)

% check whether we are inside the fundamental region
sR.checkInside(h)

% define some crystal plane
h = Miller({1,0,-1,0},{0,0,0,1},cs,'HKIL')
h = h.project2FundamentalRegion

% generate vectors within a spherical region
r = vector3d.rand(sR)
r = equispacedS2Grid(sR)
```

Crystal Orientation

Definition

Let

- $\vec{X}$, $\vec{Y}$, $\vec{Z}$ - specimen fixed orthonormal coordinate system
- $\vec{x}$, $\vec{y}$, $\vec{z}$ - crystal fixed orthonormal coordinate system

Then the basis transformation or passive rotation $\mathbf{O}$ that translates crystal coordinates $(a, b, c)^T$ of a direction

$$\vec{v} = a\vec{x} + b\vec{y} + c\vec{z}$$

into specimen coordinates $(A, B, C)^T = \mathbf{O}(a, b, c)^T$ satisfying

$$\vec{v} = A\vec{X} + B\vec{Y} + C\vec{Z}$$

is called **orientation** of the crystal lattice.

Defining Orientations

by Euler angles:

```
ori = orientation.byEuler(10*degree,20*degree,30*degree,cs)
```

by specific directions

```
n = Miller(1,1,1,'hkl',cs)
d = Miller(1,-1,0,'uvw',cs)
ori = orientation.map(n, vector3d.Z, d, vector3d.X)
```

by Miller indices $(hkl)[uvw]$:

```
ori = orientation.byMiller(n,d)
```

by rotation matrix

```
ori = orientation.byMatrix(eye(3),cs)
```

import from file

```
ori = orientation.load(fname,'Columns',{'phi1','Phi','phi2'},cs)
```

Crystal Orientations as Coordinate Transforms

```
cs    = crystalSymmetry('622')
euler = [10,20,30]*degree;
ori   = orientation.byEuler(euler,cs)
h     = Miller(1,1,-2,0,cs)
```

```
ori = orientation (622 -> xyz)

  Bunge Euler angles in degree
  phi1   Phi  phi2  Inv.
    10    20    30     0

h = Miller (622)
  h   k   i   l
  1   1  -2   0
```

Crystal Orientations as Coordinate Transforms

```
cs    = crystalSymmetry('622')
euler = [10,20,30]*degree;
ori   = orientation.byEuler(euler,cs)
h     = Miller(1,1,-2,0,cs)
```

```
ori * h
```

```
ans = vector3d
          x         y         z
   0.702179   1.77652  0.592396
```

Crystal Orientations as Coordinate Transforms

```
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euler = [10,20,30]*degree;
ori   = orientation.byEuler(euler,cs)
h     = Miller(1,1,-2,0,cs)
```

```
ori * h
```

```
inv(ori)
```

```
ans = orientation (xyz -> 622)

  Bunge Euler angles in degree
  phi1   Phi  phi2  Inv.
   150    20   170     0
```

Crystal Orientations as Coordinate Transforms

```
cs    = crystalSymmetry('622')
euler = [10,20,30]*degree;
ori   = orientation.byEuler(euler,cs)
h     = Miller(1,1,-2,0,cs)
```

```
ori * h
```

```
inv(ori) * vector3d.Z
```

```
ans = Miller (622)
  h      k       i      l
  0 0.2962 -0.2962 0.9397
```

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```

```
ori * h
```

```
inv(ori) * vector3d.Z
```

```
round(inv(ori) * vector3d.Z)
```

```
ans = Miller (622)
  h  k  i  l
  0  3 -3 10
```

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```

```
ori * h
```

```
inv(ori) * vector3d.Z
```

```
round(inv(ori) * vector3d.Z)
```

Multiplication with an <u>`orientation`</u> transforms an object from crystal into specimen coordinates. Multiplication with the inverse transforms it back.

- `vector3d`, `Miller`
- `slipSystem`, `dislocationSystem`
- `tensor`

	MTEX Orientations	**Bunge Orientations**
orientation	crystal to specimen coordinates	specimen to crystal coordinates
Euler angles	$(\varphi_1, \Phi, \varphi_2) = \mathbf{R}_{\vec{Z},\varphi_1}\mathbf{R}_{\vec{X},\Phi}\mathbf{R}_{\vec{Z},\varphi_2}$	$(\varphi_1, \Phi, \varphi_2) = \mathbf{R}_{\vec{Z}'',\varphi_2}\mathbf{R}_{\vec{X}',\Phi}\mathbf{R}_{\vec{Z},\varphi_1}$
Miller indices	✓	✓
matrix	$\mathbf{M}^{-1}$	$\mathbf{M}$
misorientation	$\mathtt{ori}_1^{-1} * \mathtt{ori}_2$	$\mathtt{ori}_1 * \mathtt{ori}_2^{-1}$
pole figure	✓	✓
ODF section	✓	✓
Import	✓	✓

An Orthogonal Reference System

We will describe the orientation of a crystal lattice within a specimen by a basis transformation from a crystal fixed coordinate system to a specimen fixed coordinate system.

What do we have so far:

- a direct crystal coordinate system $\vec{a}, \vec{b}, \vec{c}$
- a reciprocal coordinate system $\vec{a}^*, \vec{b}^*, \vec{c}^*$

In order that the basis transformation is an orthogonal matrix we need an orthogonal coordinate system $(\vec{x}, \vec{y}, \vec{z})$ fixed to the crystal lattice.

There are different conventions:

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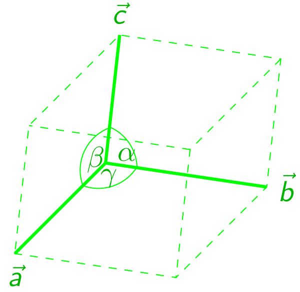

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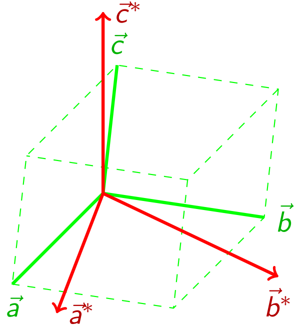

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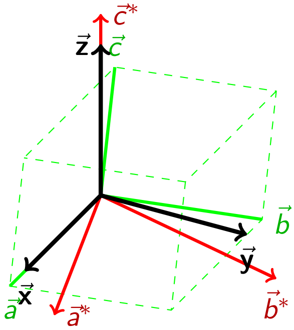

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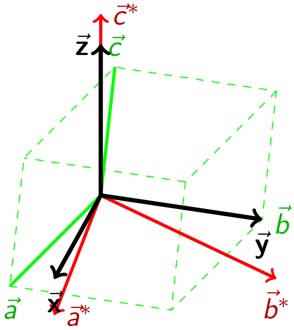

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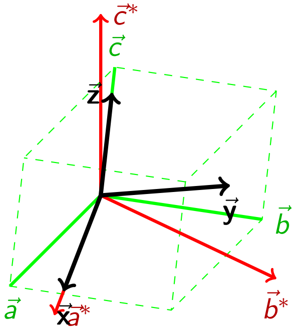
An Orthogonal Reference System

We will describe the orientation of a crystal lattice within a specimen by a basis transformation from a crystal fixed coordinate system to a specimen fixed coordinate system.

What do we have so far:

- a direct crystal coordinate system $\vec{a}, \vec{b}, \vec{c}$
- a reciprocal coordinate system $\vec{a}^*, \vec{b}^*, \vec{c}^*$

In order that the basis transformation is an orthogonal matrix we need an orthogonal coordinate system $(\vec{x}, \vec{y}, \vec{z})$ fixed to the crystal lattice.



There are different conventions:

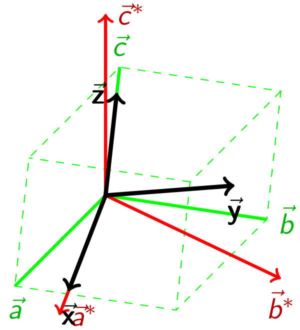
- $\vec{x}||\vec{a}$, $\vec{z}||\vec{c}^*$
- $\vec{y}||\vec{b}$, $\vec{z}||\vec{c}^*$
- $\vec{x}||\vec{a}^*$, $\vec{z}||\vec{c}$

An Orthogonal Reference System

We will describe the orientation of a crystal lattice within a specimen by a basis transformation from a crystal fixed coordinate system to a specimen fixed coordinate system.

What do we have so far:

- a direct crystal coordinate system $\vec{a}, \vec{b}, \vec{c}$
- a reciprocal coordinate system $\vec{a}^*, \vec{b}^*, \vec{c}^*$



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There are different conventions:

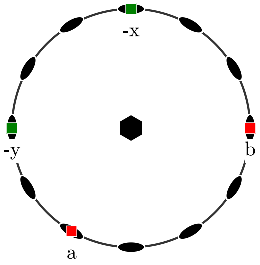
- $\vec{x}||\vec{a}$, $\vec{z}||\vec{c}^*$
- $\vec{y}||\vec{b}$, $\vec{z}||\vec{c}^*$
- $\vec{x}||\vec{a}^*$, $\vec{z}||\vec{c}$

The used conventions effects orientations and tensorial properties and should always be reported!

Defining the Orthogonal Reference Frame in MTEX

```
cs1 = crystalSymmetry('622')
```

```
cs1 = crystalSymmetry
  symmetry         : 622
  elements         : 12
  a, b, c          : 1, 1, 1
  reference frame: X||a*, Y||b, Z||c*
```



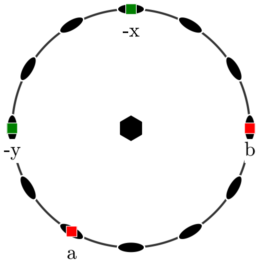
Defining the Orthogonal Reference Frame in MTEX

```
cs1 = crystalSymmetry('622')
```

```
ori = orientation.rand(cs1)
```

```
ori = orientation (622 -> xyz)
  crystal symmetry : 622, X||a*, Y||b, Z||c*

  Bunge Euler angles in degree
     phi1      Phi     phi2     Inv.
  131.348 74.3274 347.732        0
```



Defining the Orthogonal Reference Frame in MTEX

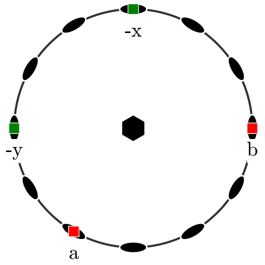
```
cs1 = crystalSymmetry('622')
```

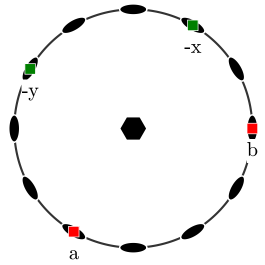
```
ori = orientation.rand(cs1)
```

```
cs2 = crystalSymmetry('622', 'x||a')
ori.CS = cs2
```

```
ori = orientation (622 -> xyz)
  crystal symmetry : 622, X||a, Y||b*, Z||c

  Bunge Euler angles in degree
     phi1      Phi     phi2     Inv.
  131.348 74.3274 347.732        0
```





Defining the Orthogonal Reference Frame in MTEX

```
cs1 = crystalSymmetry('622')
```

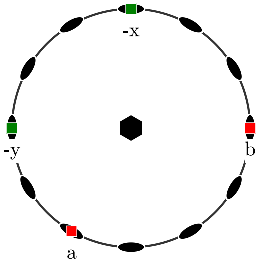
```
ori = orientation.rand(cs1)
```

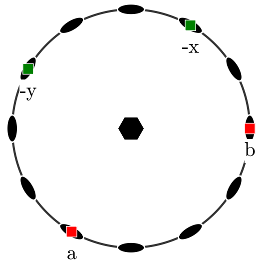
```
cs2 = crystalSymmetry('622', 'x||a')
ori.CS = cs2
```

```
ori2 = ori.transformReferenceFrame(cs2)
```

```
ori = orientation (622 -> xyz)
  crystal symmetry : 622, X||a, Y||b*, Z||c

  Bunge Euler angles in degree
     phi1      Phi     phi2    Inv.
  131.348 74.3274  317.732       0
```





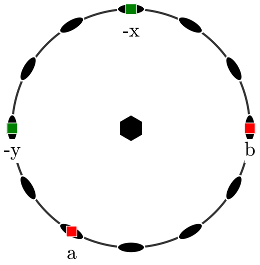
Defining the Orthogonal Reference Frame in MTEX

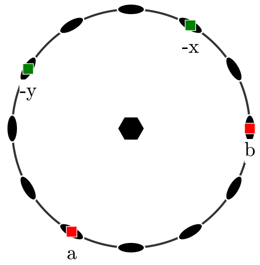
```
cs1 = crystalSymmetry('622')
```

```
ori = orientation.rand(cs1)
```

```
cs2 = crystalSymmetry('622','x||a')
ori.CS = cs2
```

```
ori2 = ori.transformReferenceFrame(cs2)
```





the command `transformReferenceFrame`

- is available for objects of type `Miller`, `tensor`, `slipSystem`, `dislocationSystem`, `ODF`
- does not change orientation
- is automatically called if an operation incorporates different setups

Symmetrically Equivalent Orientations

Two orientations $\mathbf{O}_1$, $\mathbf{O}_2$ are symmetrically equivalent if there is a symmetry $\mathbf{S} \in \mathcal{S}$ with

$$\mathbf{O}_2 = \mathbf{O}_1 \mathbf{S}.$$

- We may either symmetrise crystal directions or orientations.
- The misorientation angle between symmetrically equivalent orientations is 0.
- The crystallographic equivalent orientations form a regular body in $\mathbb{R}^4$.
- We may restrict the class of symmetrically equivalent orientations to a unique representative.

```
ori.symmetrise
```

```
ans = orientation (622 -> xyz)
  size: 12 x 1

  Bunge Euler angles in degree
  phi1  Phi  phi2  Inv.
    10    20    30     0
   190   160   210     0
    10    20    90     0
   190   160   270     0
    10    20   150     0
   190   160   330     0
    10    20   210     0
   190   160    30     0
    10    20   270     0
   190   160    90     0
    10    20   330     0
   190   160   150     0
```

Symmetrically Equivalent Orientations

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- The crystallographic equivalent orientations form a regular body in $\mathbb{R}^4$.
- We may restrict the class of symmetrically equivalent orientations to a unique representative.

```
ori.symmetrise
```

```
ori.symmetrise * h
ori * h.symmetrise
ori * cs * h
```

```
ans = vector3d
 size: 12 x 1
          x         y         z
    0.89058   0.70798    0.1974
    0.89058   0.70798    0.1974
   -0.18849    1.0688    0.3949
     1.0792  -0.36061   -0.1974
    -1.0792   0.36061    0.1974
    0.18849   -1.0688   -0.3949
   -0.89058  -0.70798   -0.1974
   -0.89058  -0.70798   -0.1974
    0.18849   -1.0688   -0.3949
    -1.0792   0.36061    0.1974
     1.0792  -0.36061   -0.1974
   -0.18849    1.0688    0.3949
```

Symmetrically Equivalent Orientations

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- We may restrict the class of symmetrically equivalent orientations to a unique representative.

```
ori.symmetrise
```

```
ori.symmetrise * h
ori * h.symmetrise
ori * cs * h
```

```
angle(ori, ori * cs)
```

```
ans =
   0
   0
   0
   0
   0
   0
   0
   0
   0
   0
   0
   0
```

Symmetrically Equivalent Orientations

Two orientations $\mathbf{O}_1$, $\mathbf{O}_2$ are symmetrically equivalent if there is a symmetry $\mathbf{S} \in \mathcal{S}$ with

$$\mathbf{O}_2 = \mathbf{O}_1 \mathbf{S}.$$

- We may either symmetrise crystal directions or orientations.
- The misorientation angle between symmetrically equivalent orientations is 0.
- The crystallographic equivalent orientations form a regular body in $\mathbb{R}^4$.
- We may restrict the class of symmetrically equivalent orientations to a unique representative.

```
ori.symmetrise
```

```
ori.symmetrise * h
ori * h.symmetrise
ori * cs * h
```

```
angle(ori, ori * cs)
```

```
angle(ori, ori * cs,...
  'noSymmetry')./degree
```

```
ans =
     0.0000
   180.0000
    60.0000
   180.0000
   120.0000
   180.0000
   180.0000
   180.0000
   120.0000
   180.0000
    60.0000
   180.0000
```

Symmetrically Equivalent Orientations

Two orientations $\mathbf{O}_1$, $\mathbf{O}_2$ are symmetrically equivalent if there is a symmetry $\mathbf{S} \in \mathcal{S}$ with

$$\mathbf{O}_2 = \mathbf{O}_1 \mathbf{S}.$$

- We may either symmetrise crystal directions or orientations.
- The misorientation angle between symmetrically equivalent orientations is 0.
- The crystallographic equivalent orientations form a regular body in $\mathbb{R}^4$.
- We may restrict the class of symmetrically equivalent orientations to a unique representative.

```
ori.symmetrise
```

```
ori.symmetrise * h
ori * h.symmetrise
ori * cs * h
```

```
angle(ori, ori * cs)
```

```
angle(ori, ori * cs,...
  'noSymmetry')./degree
```

```
ori.project2FundamentalRegion
```

```
ans = orientation (622 -> xyz)

  Bunge Euler angles in degree
  phi1   Phi  phi2  Inv.
    10    20   330     0
```

Fundamental Regions of the Orientation Space

Definition

The fundamental region is a region in the space of all rotations which contains from each class of symmetrically equivalent rotations exactly one.

- The fundamental region is no uniquely defined.
- Its volume is always $\frac{1}{|\mathcal{S}|}$ of the entire rotation space.

The fundermental region in Euler angle space

symmetry	1	112	222	3	321	4	422	6	622	23	432
φ_1	360	360	360	360	360	360	360	360	360	360	360
Φ	180	180	90	180	90	180	90	180	90	90	90
φ_2	360	180	180	120	120	90	90	60	60	180	90

In cubic symmetry the three fold axis is not respected.

Fundamental Regions as Voronoi Cells

We may use the rotational angle $\omega(\mathbf{O}, \tilde{\mathbf{O}})$ to define Voronoi cells in orientation space.

Definition

Let $\mathbf{O}$ be an arbitrary orientation with symmetry group $\mathcal{S}$. The Voronoi cell $\mathcal{V}_\mathbf{O}$ is defined as all orientations $\tilde{\mathbf{O}}$ which have smaller distance to $\mathbf{O}$ then to any of its symmetrically equivalent orientations $\mathbf{OS}$, $\mathbf{S} \in \mathcal{S}$, i.e.,

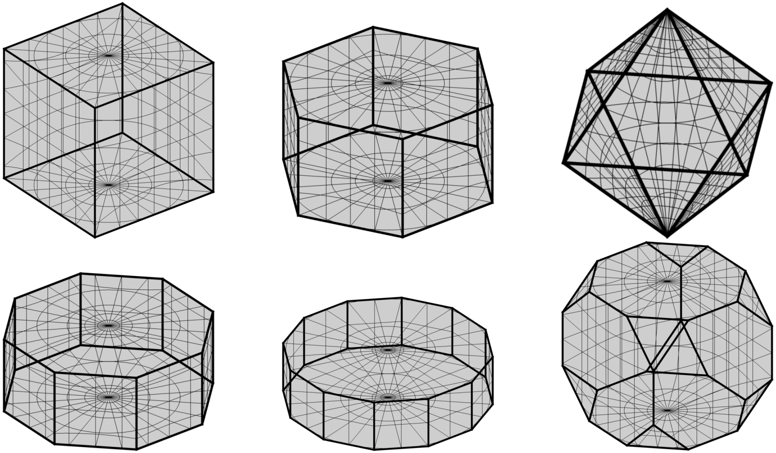
$$\mathcal{V}_\mathbf{O} = \left\{ \tilde{\mathbf{O}} \text{ such that for all } \mathbf{S} \in \mathcal{S} \; \omega(\tilde{\mathbf{O}}, \mathbf{O}) \leq \omega(\tilde{\mathbf{O}}, \mathbf{OS}) \right\}$$

Theorem

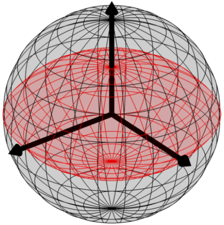
The Voronoi cell $\mathcal{V}_\mathbf{O}$ is a fundamental region which is bounded by the midplanes between $\mathbf{O}$ and all its symmetrically equivalent orientations $\mathbf{OS}$, $\mathbf{S} \in \mathcal{S}$.

Fundamental Regions in Rodrigues Space

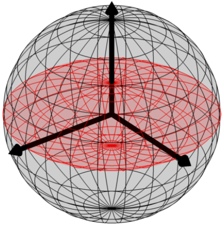
Since in Rodrigues Frank space the midplanes between two orientations are “real” Euclidean planes the bounded Voronoi cells are regular polyhedrons.



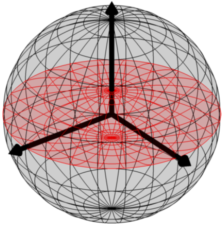
Fundamental Regions in Axis Angle Space



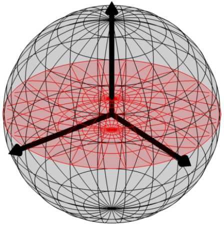
2



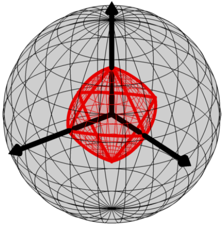
3



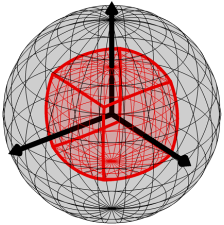
4



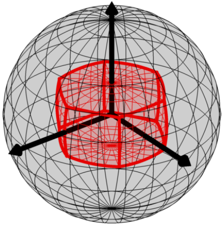
6



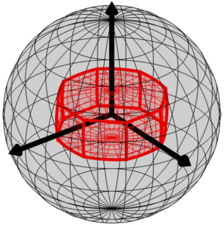
23



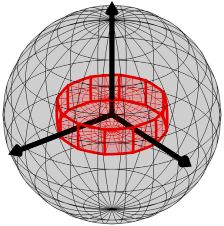
222



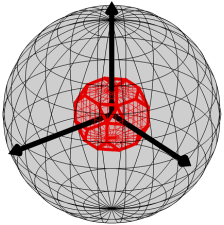
321



422



622



432

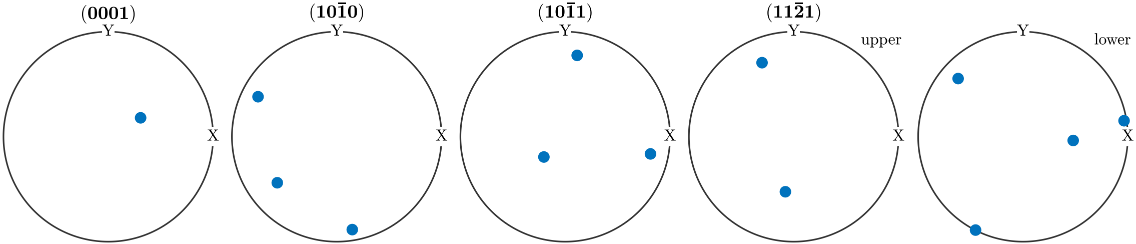
Pole Figures (crystal directions in specimen coordinates)

- $\vec{h}$ - crystal direction
- $\mathbf{S}\vec{h}$ - crystallographic equivalent directions
- $\mathbf{O}$ - orientation

$\Longrightarrow$ crystallographic equivalent directions in specimen coordinates: $\mathbf{O}\mathbf{S}\vec{h}, \quad \mathbf{S} \in \mathcal{S}$

Spherical plots of these directions are called pole figures.

- every orientation appears $|\mathcal{S}\vec{h}|$ times
- if $\vec{h}$ and $-\vec{h}$ are symmetric only the upper hemisphere needs to be plotted
- pole figures reflects the macroscopic symmetries of the specimen



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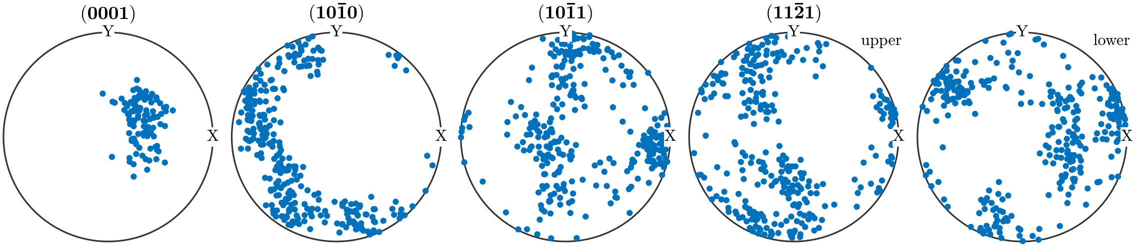


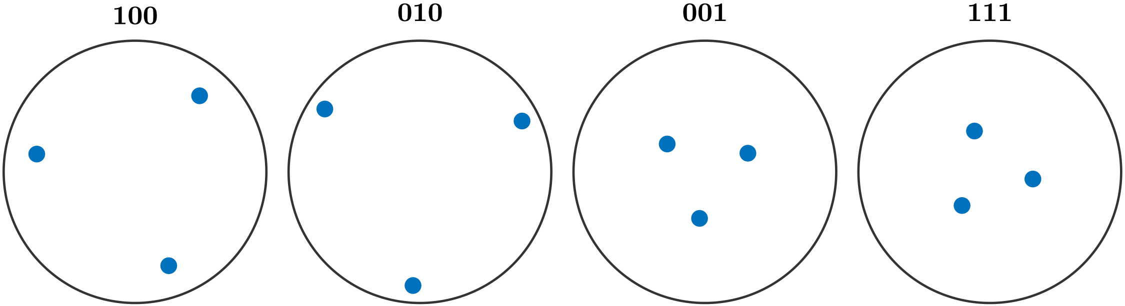
Inverse Pole Figures (specimen directions in crystal coordinates)

- $\vec{r}$ - specimen direction
- $\mathbf{O}$ - orientation

$\Longrightarrow$ crystal directions pointing in direction $\vec{r}$: $\mathbf{S}\mathbf{O}^{-1}\vec{r}, \quad \mathbf{S} \in \mathcal{S}$

Spherical plots of these directions are called inverse pole figures.

- every orientation appears $|\mathcal{S}|$ times



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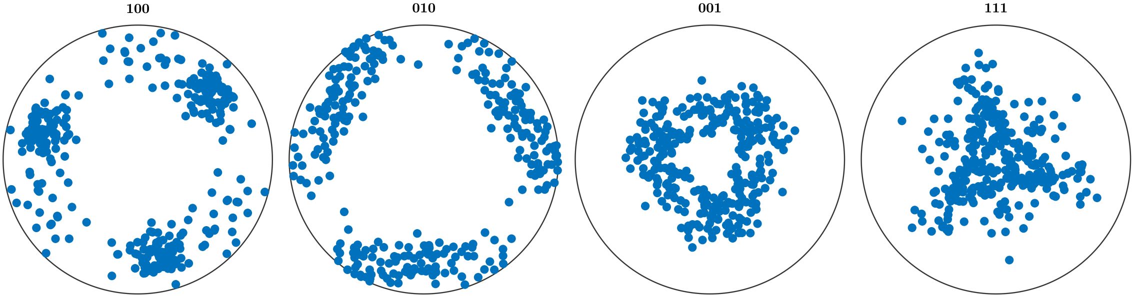

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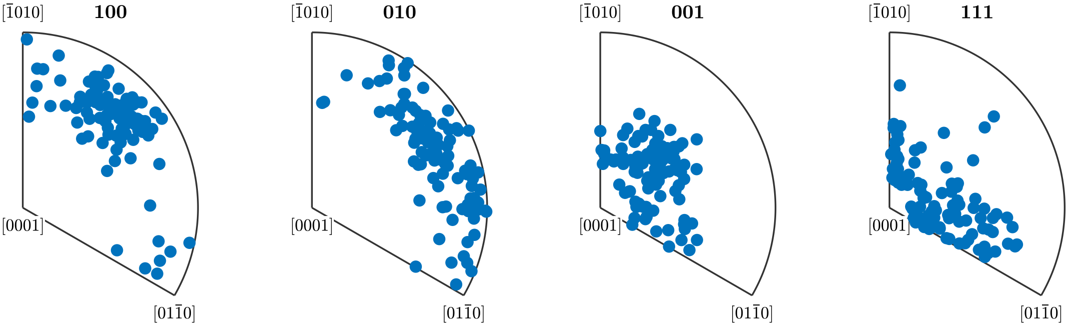
$\Longrightarrow$ crystal directions pointing in direction $\vec{r}$: $\mathbf{S}\mathbf{O}^{-1}\vec{r}, \quad \mathbf{S} \in \mathcal{S}$

Spherical plots of these directions are called inverse pole figures.

- every orientation appears $|\mathcal{S}|$ times
- only the fundamental sector needs to be plotted



MTEX

```
cs = crystalSymmetry.load('quartz');
ori = orientation.rand(100,cs);

% pole figure plots
h = Miller(1,1,-2,0,cs)
plot(ori * h.symmetrise)
plotPDF(ori,h)

% inverse pole figure plots
r = vector3d.Z
plot(inv(ori.symmetrise) * r)
plotIPDF(ori,r)
```

Visualizing Orientations

3d plots

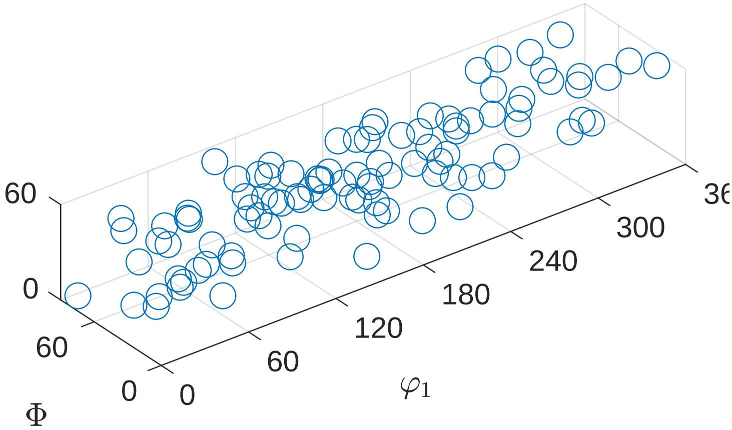
- Euler angle space
- axis angle space
- Rodrigues space

section plots

- Euler angle sections
- axis angle sections
- sigma sections

projection plots

- pole figures
- inverse pole figures



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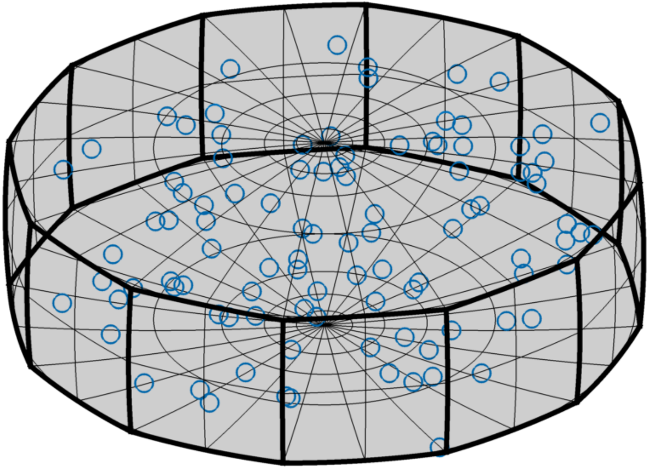

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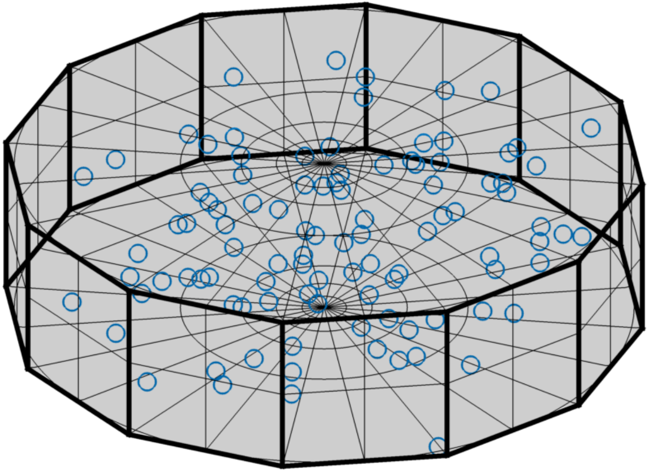

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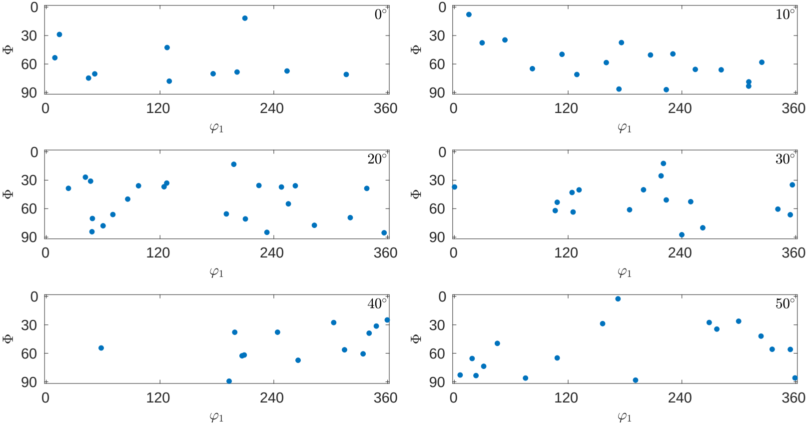

Visualizing Orientations

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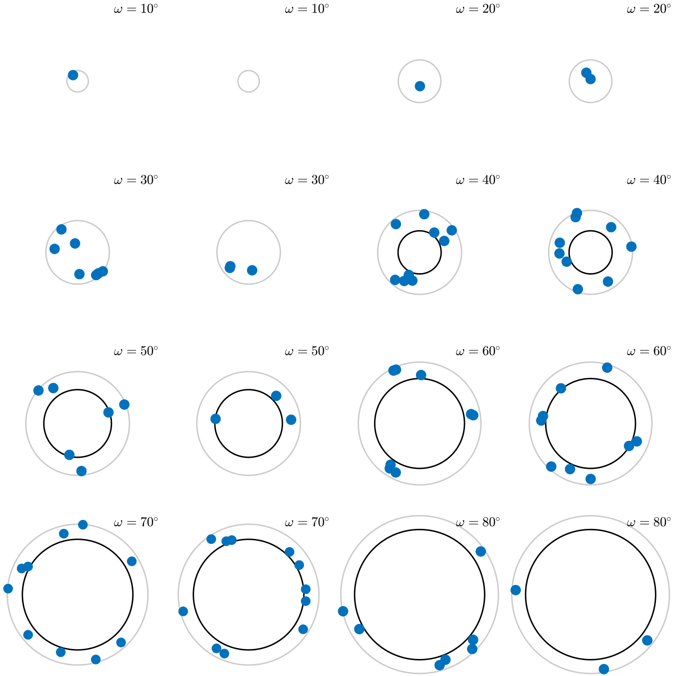

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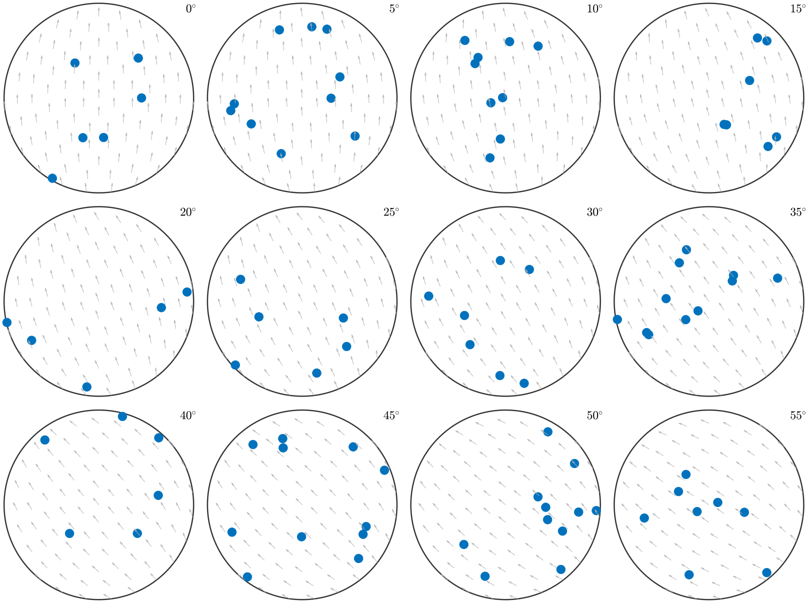

Visualizing Orientations

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section plots

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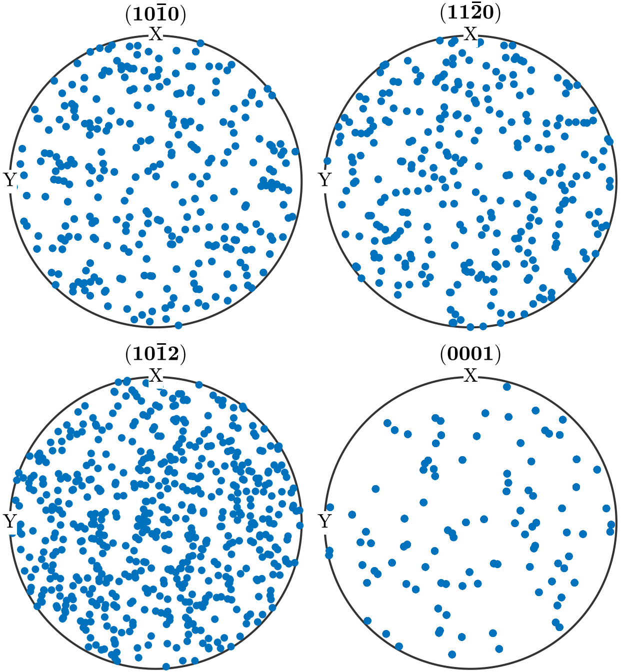

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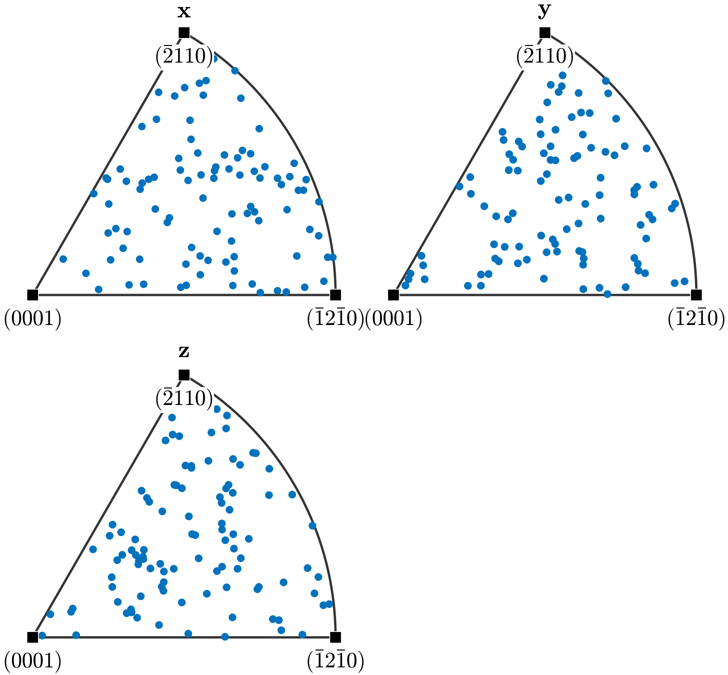


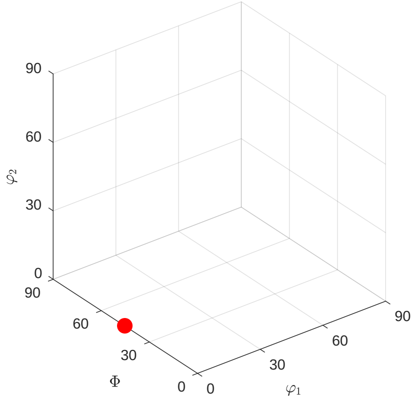
MTEX

```
% some random orientations
cs = crystalSymmetry('cubic')
ori = orientation.rand(100,cs)

% 3d plots
plot(ori) % Euler angle plots
plot(ori,'axisAngle')
plot(ori,'Rodrigues')

% sections
plotSection(ori,'phi2',(0:10:50)*degree)
plotSection(ori,'sigma','sections',12)
plotSection(ori,'axisAngle')
```

Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

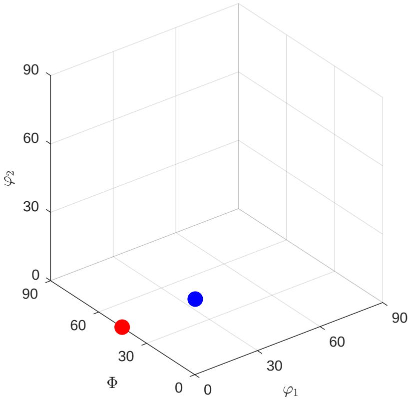


Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

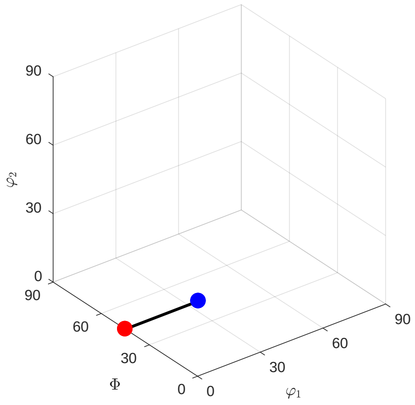
The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
f = fibre(o1,o2)
```

```
f = fibre (432 -> xyz (222))

 h || r: (011) || (0,0,1)
 o1 -> o2: (0°,45°,0°) -> (35°,45°,0°)
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

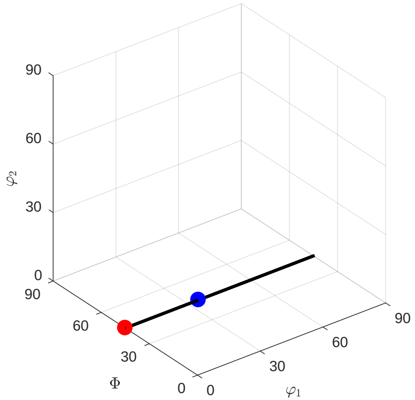
The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
f = fibre(o1,o2,'full')
```

```
f = fibre (432 -> xyz (222))

 h || r: (011) || (0,0,1)
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

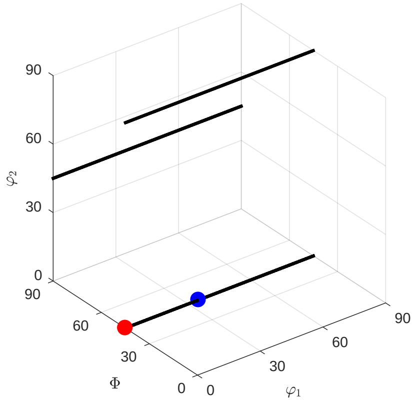
```
f = fibre(o1,o2,'full')
```

```
f.symmetrise
```

```
f = fibre (show methods, plot)
  size: 12 x 1
  crystal symmetry: 432 specimen
  symmetry: 222
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

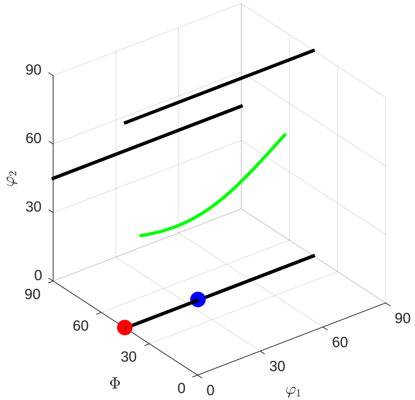
```
f = fibre(o1,o2,'full')
```

```
f.symmetrise
```

```
f = fibre.beta(cs,ss)
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
f = fibre(o1,o2,'full')
```

```
f.symmetrise
```

```
f = fibre.beta(cs,ss)
```

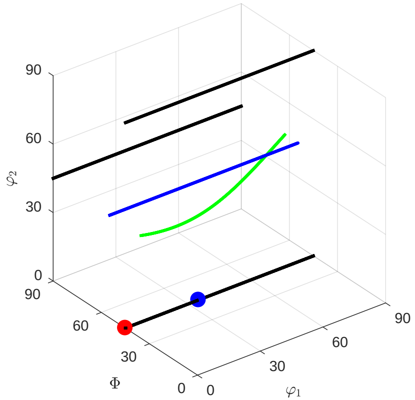
The full fiber is characterized by a crystal direction $(hk\ell)$ and a specimen direction (xyz) as all orientations `ori` satisfying

$$\mathtt{ori} * (hk\ell) = (xyz)$$

```
fibre(Miller(1,1,1,cs),zvector)
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
f = fibre(o1,o2,'full')
```

```
f.symmetrise
```

```
f = fibre.beta(cs,ss)
```

The full fiber is characterized by a crystal direction $(hk\ell)$ and a specimen direction (xyz) as all orientations `ori` satisfying

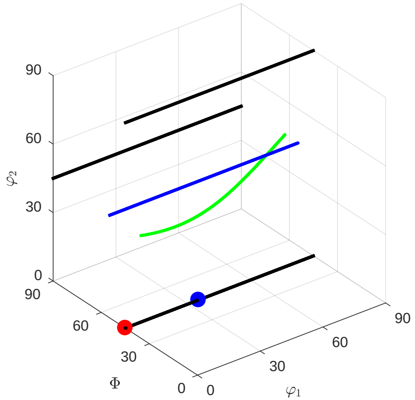
$$\texttt{ori} * (hk\ell) = (xyz)$$

```
fibre(Miller(1,1,1,cs),zvector)
```

```
f = fibre.fit(ori)
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```



Fibers of Orientations

The shortest line between two orientations `ori1` and `ori2` is called fibre.

```
f = fibre(o1,o2,'full')
```

```
f.symmetrise
```

```
f = fibre.beta(cs,ss)
```

The full fiber is characterized by a crystal direction $(hk\ell)$ and a specimen direction (xyz) as all orientations `ori` satisfying

$$\mathtt{ori} * (hk\ell) = (xyz)$$

```
fibre(Miller(1,1,1,cs),zvector)
```

```
f = fibre.fit(ori)
```

```
angle(ori,f)
```

```
cs = crystalSymmetry('432')
ss = specimenSymmetry('222')
o1 = orientation.goss(cs,ss)
```

```
o2 = orientation.brass(cs,ss)
```